Check for updates

Transposable elements shape the evolution of mammalian development

Anna D. Senft ✉ *and Todd S. Macfarlan* ✉

Abstract | Transposable elements (TEs) promote genetic innovation but also threaten genome stability. Despite multiple layers of host defence, TEs actively shape mammalian-specific developmental processes, particularly during pre-implantation and extra-embryonic development and at the maternal–fetal interface. Here, we review how TEs influence mammalian genomes both directly by providing the raw material for genetic change and indirectly via co-evolving TE-binding Krüppel-associated box zinc finger proteins (KRAB-ZFPs). Throughout mammalian evolution, individual activities of ancient TEs were co-opted to enable invasive placentation that characterizes live-born mammals. By contrast, the widespread activity of evolutionarily young TEs may reflect an ongoing co-evolution that continues to impact mammalian development.

Transposable elements (TEs). Mobile DNA segments present in virtually all domains of life whose remains make up nearly 50% of most mammalian genomes.

Transpose
Pertaining to transposition, which is the process by which transposable elements (TEs) change their genomic position, leading to the description of TEs as 'jumping genes'.

Krüppel-associated box zinc finger proteins
(KRAB-ZFPs). KRAB domain containing C2H2-type zinc finger proteins that predominantly act as transcriptional repressors of transposable elements (TEs).

Less than 2% of the human genome encodes genes[1]. They are to a large extent identical between different mammals: we share >98% of genes with chimpanzees and >85% with mice[2,3]. Evolution thus relies largely on changes in non-coding regions of genomes, particularly in regulatory sequences that alter gene expression[4]. The majority of the genetic information in humans (and most mammals) consists of transposable elements (TEs)[1,5,6]. Intriguingly, regulatory sequences often originate from TEs[7–11]. TEs are repetitive genetic sequences that once had or still have the ability to transpose, that is, to mobilize and insert elsewhere in the genome. In contrast to genes, TEs are enormously diverse across species and often species-specific[5].

TEs have diverse origins and can be classified based on their mode of transposition (BOX 1). Even though most mammalian TEs no longer transpose, the continuous genomic infiltration, propagation and degradation of TEs over time have promoted genetic innovation on an evolutionary scale[5,12]. On the other hand, the repetitive nature of TEs can threaten genomic stability, and uncontrolled transposition contributes to disease[13]. The colonization of genomes with TEs has therefore led to the evolution of diverse defence systems that limit their mobilization. These defence systems include, among others, DNA modifications, small RNA silencing pathways and sequence-specific transcription factors[14–16]. A prominent vertebrate TE defence system is TE-binding Krüppel-associated box zinc finger proteins (KRAB-ZFPs) that use C2H2-type zinc finger arrays to bind DNA, most often within TEs[17] (BOX 2). Although KRAB-ZFPs are abundant throughout most vertebrates, live-born mammals underwent a burst in the evolution of KRAB-ZFP genes, many of which endured through purifying selection[17] (FIG. 1). Consequently, KRAB-ZFPs have become the largest transcription factor family in humans and mice[18,19]. In vertebrate genomes, there is a striking correlation between the number of host tandem zinc finger genes (which include many KRAB-ZFPs) and a specific class of TEs, long terminal repeat (LTR) retroelements[18]. In mammals, most LTR retroelements are endogenous retroviruses (ERVs), which derive from ancient and ongoing retroviral infections of the germ line (BOX 1). The LTRs of ERVs can recombine and are thus often found individually as solo LTRs. Although ERVs/LTR elements are also found in other vertebrates, they are prevalent in the genomes of all live-born mammals and not prevalent in the genome of the egg-laying platypus[6] (FIG. 1a).

The predecessors of mammals were reptiles that, like mammals and birds, belong to the animal group of amniotes. Amniotes evolved partly through the diversification of extra-embryonic tissues[20]. Approximately 180 million years ago (Ma) these extra-embryonic tissues evolved further to become an integral part of the mammalian placenta, leading to the emergence of mammals. The first mammals, the monotremes (for example, platypus and echidna), still laid eggs. By contrast, marsupials (for example, kangaroo and opossum) and eutheria (for example, mouse and human) that diverged ~160 Ma from a common therian ancestor are live-born, enabled by a co-evolution of extra-embryonic tissues with maternal tissues that gave rise to an invasive placenta (BOX 3). Here, we summarize evidence that these evolutionary changes are deeply connected with TEs and TE-binding KRAB-ZFPs. TEs fundamentally contribute to the development and evolution of mammals by providing species-specific regulatory elements and novel proteins. Moreover, TE-binding KRAB-ZFPs adopted roles in TE regulation and, in some cases, acquired novel functions essential for mammalian development.

The Eunice Kennedy Shriver National Institute of Child Health and Human Development, The National Institutes of Health, Bethesda, MD, USA.

✉*e-mail: anna.senft@nih.gov; todd.macfarlan@nih.gov*

https://doi.org/10.1038/s41576-021-00385-1

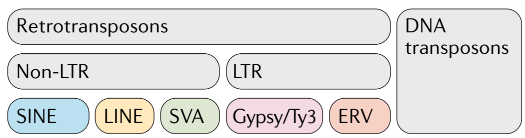
Box 1 | Classification, origins and activity of TEs

When were TEs discovered?

Barbara McClintock discovered transposable elements (TEs) in the 1950s by attributing pigment changes in maize to TE transposition[242]. McClintock also first proposed that TEs, or controlling elements, as she called them, regulate development[243].

How are TEs classified?

Broadly, TEs can be divided into two classes: DNA transposons excise and insert a DNA intermediate when they transpose ('cut and paste'), whereas retrotransposons reverse transcribe RNA intermediates prior to integration ('copy and paste'). DNA transposons are few and inactive in most mammals whereas retrotransposons are abundant[244]. Retrotransposons are further classified by whether they contain long terminal repeats (LTRs). Most LTR-containing retrotransposons in mammals are endogenous retroviruses (ERVs). Frequent recombination between LTRs leaves behind many solo LTRs in the genome. Retrotransposons lacking LTRs include autonomous long interspersed elements (LINEs) and non-autonomous short interspersed elements (SINEs), which require LINE-derived proteins for their mobilization. In addition to LINEs and SINEs, humans encode additional primate-specific composite elements called SINE variable-number tandem-repeat Alu (SVA) elements (see the figure). The specific TE examples discussed in the main text only represent a small fraction of the TEs found in mammals. To find out more about mammalian TE classification, see REF.[5] and visit Dfam TE Classification and TEhub (see Related links).



Where do TEs come from?

The origin of most TEs is uncertain[5]. Whereas ERVs are likely to have arisen from ancient viral infections, some non-LTR retrotransposons may have evolved from self-splicing group II introns in bacteria[245]. These group II intron TE predecessors are still mobile in eukaryote organelles and gave rise to the spliceosome, thereby contributing to eukaryote evolution[245–247]. Many SINEs arose from cellular RNAs such as tRNAs[248]. Whereas SINEs evolved several times independently, LINEs are all related to each other and can be traced back to eukaryotes[248,249]. RNA-binding domains of diverse TE-encoded proteins even occur in all cellular life[250].

Do mammalian TEs still jump?

Although in most mammals DNA transposons and the majority of retrotransposons are inactive, some copies of distinct retrotransposon families can still mobilize including LINEs (L1; also known as LINE-1), SINEs (B1, B2) and ERVs (intracisternal A-type particles (IAPs) and early transposons (ETns)) in mice and L1 and Alu in humans[251–255]. Occasionally, TEs spread to other species and can transmit disease; for example, as membrane-enclosed virions in the case of koala ERVs. This horizontal transfer blurs the distinction between exogenous retroviruses and ERVs[256,257].

C2H2-type zinc finger
One of the most common domains found in the transcription factors of higher eukaryotes consisting of two cysteines and two histidines that coordinate the binding of a zinc ion, most frequently found in an array of repeated fingers encoded by a single exon.

Purifying selection
Preservation of a nucleotide or amino acid sequence by selective removal of mutations that reduce fitness (for example, visible by an excess of synonymous mutations, which conserve protein structure, relative to non-synonymous mutations).

Intriguingly, despite TE silencing mechanisms such as KRAB-ZFPs, TEs are also dynamically expressed throughout mammalian development, often with unclear effects[21–23]. Although their expression could reflect an 'accidental' by-product of stage and tissue-specific epigenetic patterns, accumulating evidence suggests that TEs contribute actively to gene regulation in a highly species, cell type and stage-specific manner[24–28]. In fact, TE expression patterns are so specific that they can stage embryos, determine evolutionary relationships, label cancer cell types, predict disease outcome and characterize neurodegenerative disease[10,25,27–36]. Moreover, the specific expression of TEs in disease may be a direct consequence of their developmental functions in live-born mammals.

Since their discovery, TEs have been hard to study, as they are repetitive, short sequences that are difficult to map and analyse, and have thus often been excluded from genome-wide studies. Their sheer abundance has been impossible to tackle with classical gene targeting approaches. Only recently, improved long-read sequencing and genome editing technologies and a wealth of novel analysis and annotation tools have enabled the genome-wide analysis of TEs and TE-regulating mechanisms[37–39]. Complementing these genetic tools, live embryo imaging at high resolution and single-cell sequencing have made mammalian embryos more accessible than ever before[40,41]. These recent developments have led to much greater appreciation of the abundant and dynamic expression patterns of TEs and have fuelled attempts to assess their possible function and interplay with host defence mechanisms, particularly during embryonic development and evolution.

In this Review, we first introduce how TEs directly and indirectly shape mammalian genomes. We then discuss how both TEs and TE-binding KRAB-ZFPs contribute to critical, often mammalian or species-specific, processes during mammalian development. Our discussion follows a developmental timeline progressing from prior to implantation, through gastrulation at the interface of embryonic and extra-embryonic tissues after implantation and, finally, to maternal–fetal interactions.

How TEs shape mammalian genomes

TEs directly shape genomes

TEs encode regulatory elements (such as promoters, transcription factor binding sites and polyadenylation signals) and proteins (such as envelopes, capsids, polymerases and transposases)[12]. Once integrated into the host genome, TEs replicate within the host and are subject to natural selection. Selection of TEs for novel host functions leads to the evolution of proteins and genes, non-coding RNAs and *cis*-regulatory elements, a process called co-option[12,15]. In rare documented cases, co-opted TEs have become essential[42]. As mentioned above, TE mobilization is also mutagenic, which can have disastrous outcomes: >100 insertions of long interspersed element 1 (L1; also known as LINE-1) cause human disease by disrupting host genes, and LTRs can drive oncogenes[35,43,44]. Additionally, the repetitive nature of TEs promotes ectopic recombination and chromosomal rearrangements[45].

TEs indirectly shape genomes via KRAB-ZFPs

Humans and mice encode more than 400 KRAB-ZFPs[17] (FIG. 1b). Once bound to DNA via their C2H2-type zinc finger arrays, the KRAB domain recruits chromatin modifiers (FIG. 1c). One cofactor crucial for embryonic development and transcriptional repression downstream of KRAB-ZFPs is KRAB-associated protein 1 (KAP1) (encoded by the *Trim28* gene)[46–48]. Although not repressive by itself, KAP1 can indirectly induce heterochromatin by recruiting the trimethylated histone H3 lysine 9 (H3K9me3) methyltransferase SET (Su(var)3–9, Enhancer-of-zeste and Trithorax) domain bifurcated histone lysine methyltransferase 1 (SETDB1), which is also essential for development[49,50]. The KRAB-ZFP–KAP1 complex can further promote DNA methylation by associating with

Box 2 | **KRAB-ZFPs evolve to target TEs**

Where do KRAB-ZFPs come from?

C2H2-type zinc finger genes are found in all domains of life, but all Krüppel-associated box zinc finger proteins (KRAB-ZFPs) (which make up about half of all zinc finger genes in mice and humans) diversified from *Prdm9* (also called *Meisetz*), the oldest C2H2-type zinc finger gene with a KRAB domain[47]. A proto-form of *Prmd9* can be found in sea urchins and, thus, dates back ~520 million years[258]. But even the common ancestor of plants, fungi and animals encoded a KRAB interior motif, a potential KRAB progenitor[258]. The oldest living organism harbouring several (>200) KRAB-ZFPs is the 420 million-year-old coelacanth[17]. As coelacanths are a sister group of lungfish, KRAB-ZFPs probably became prevalent in the common ancestor of land-living vertebrates.

How do we know that KRAB-ZFPs evolve in response to TEs?

- The majority of human and mouse KRAB-ZFPs bind to specific transposable elements (TEs)[17,26,259] (FIG. 1e,f).
- Genetic ablation of many KRAB-ZFPs leads to TE reactivation[26,230].
- Many KRAB-ZFPs can be dated to a similar age as the TE groups they target[18].
- The number of host tandem zinc finger genes (which include many KRAB-ZFPs) in vertebrate species correlates with their long terminal repeat (LTR) load[18].

Do KRAB-ZFPs always completely silence TEs?

The answer to this question is no. Whereas many mammalian KRAB-ZFPs repress TEs via interacting with KRAB-associated protein 1 (KAP1) and its co-repressors (see main text), the expression of KRAB-ZFPs and their bound TEs in fact positively correlate[26,47,260]. KRAB-ZFPs may thus be regulated to 'sense' TE activation to dampen their expression. This dampening can also be context-dependent to allow tissue-specific enhancer activity of co-opted TEs[26,102]. Furthermore, very old KRAB domains, including that of PRDM9, do not interact with KAP1 (REF.[47]). Other ancient KRAB-ZFPs contain additional domains such as DUF3669 or SCAN of mostly unknown functions[261–264]. Some of the oldest KRAB-ZFPs bind mostly to gene promoters and can activate genes[17,60]. As it is impossible to determine whether these ancient promoters were once TEs, it remains unknown whether the earliest KRAB-ZFPs evolved to bind and repress TEs.

Long terminal repeat
(LTR). A direct repeat that marks the boundaries of LTR retrotransposons that contain regulatory elements and polyadenylation signals that regulate and enable viral transcription. Initially identical LTRs can recombine, giving rise to so-called solo LTRs.

Endogenous retroviruses
(ERVs). Proviral remnants of ancient retroviral infections of the germ line. ERVs encode viral proteins such as envelopes (env), polymerases (pol) and structural proteins (gag) that, in some cases, allow them to exit as viral particles.

Germ line
Cells in an organism that pass on genetic material to the progeny. The germ line comprises egg and sperm cells (germ cells) and their precursors (primordial germ cells).

Amniotes
A clade of tetrapod vertebrates that comprises reptiles, birds and mammals that evolved ~320 million years ago and relies on internal fertilization and novel extra-embryonic tissues to develop on land.

Extra-embryonic tissues
Fetus-derived tissues that support embryonic development but (for the most part) do not give rise to embryonic structures.

Placenta
A temporary, shared organ of embryonic, extra-embryonic and (in eutheria) also maternal origin that allows development inside the mother by mediating nutrient and waste exchange.

Eutheria
Mammals that develop a placenta from three tissues: the maternal decidua as well as the fetal allantois and chorion.

Therian
Live-born mammals that include marsupials and eutheria.

Invasive placenta
A placenta is invasive when it makes direct contact with maternal blood. This is the case in both mouse and humans, as trophectoderm-derived trophoblast giant cells replace maternal endothelial cells to line maternal blood vessels.

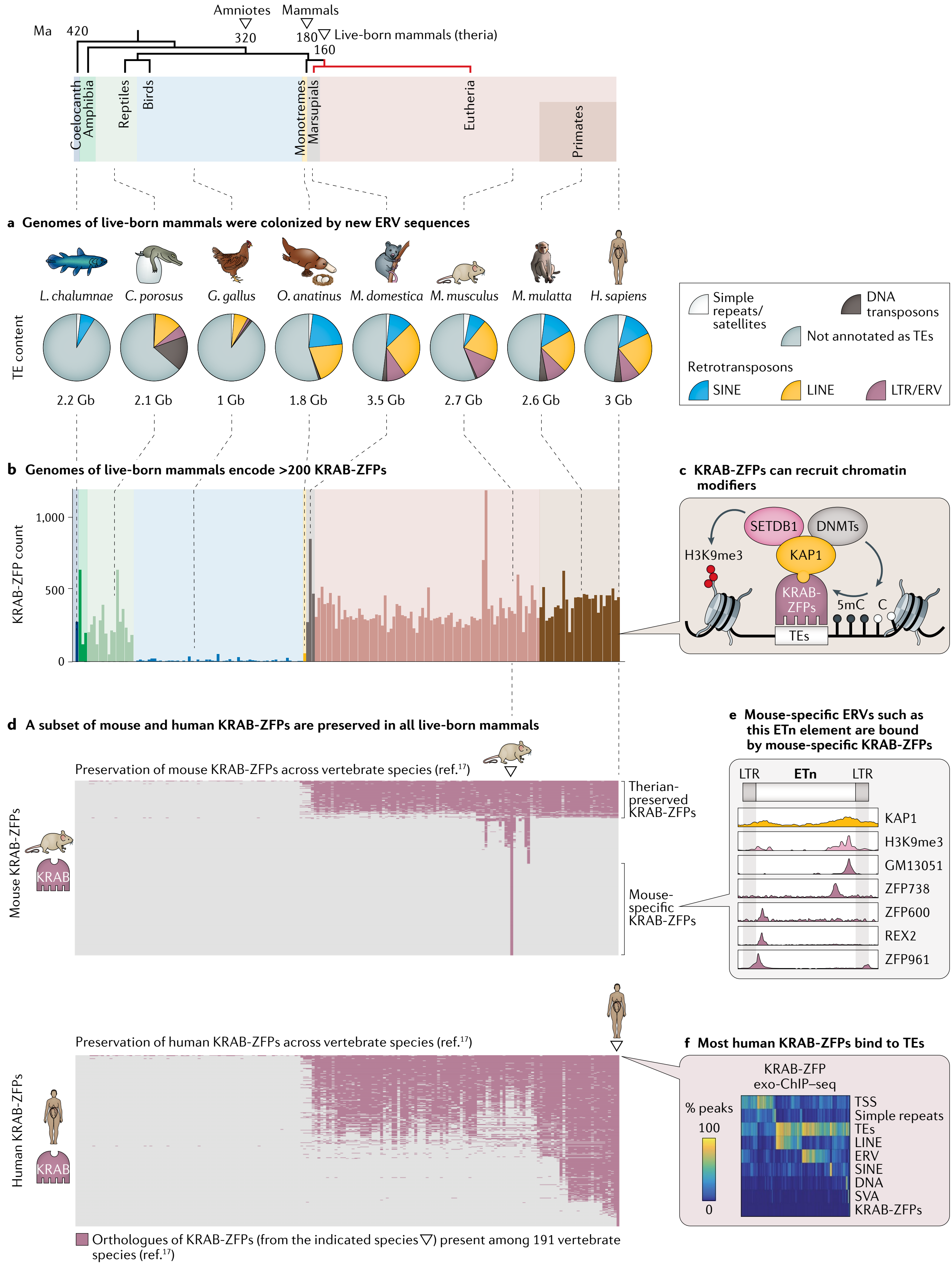
DNA methyltransferases that add a methyl group on cytosine to form 5-methyl-cytosine (5mC)[51]. DNA methylation stably silences TEs after implantation and also contributes to essential developmental processes in mammals[52–54]. Whether DNA methylation evolved to control TEs or genes (or genes via TEs) is still debated, especially as several non-mammalian animals lack DNA methylation[55–57]. In mice and humans, individual KRAB-ZFPs — such as ZFP57 and ZFP445 in mice and their orthologues ZNF57 and ZNF445 in humans — rely on binding 5mC to spread silencing via KAP1–SETDB1, yet a possibly broader interplay between the KRAB-ZFP–KAP1 system and DNA methylation remains to be explored[58,59].

As KAP1 is unable to bind DNA, KRAB-ZFPs provide the specificity needed to silence diverse TEs. Comparable with antibodies that generate specificity towards antigens through recombination and mutation, the repetitive zinc finger arrays of KRAB-ZFPs frequently recombine and acquire mutations in DNA-contacting zinc fingerprint amino acids to achieve diverse and highly specific DNA binding[26,58,60]. If targeted, TEs must evolve to escape repression in order to remain active. This evolution, along with new colonizations of TEs, pressures KRAB-ZFPs to turn over in response, leading to an 'arms race'[15,16]. This quick turnover may explain why an astonishing amount of KRAB-ZFPs are species-specific. As discussed above, a subset of KRAB-ZFPs has, however, been retained by purifying selection, suggesting that they occasionally gain non-TE-silencing functions, even if they may still bind to TEs[17,61] (FIG. 1d). Some of these conserved KRAB-ZFPs became indispensable for mammalian development[62–64]. Although we will discuss these essential 'old' KRAB-ZFPs as well as ancient co-opted TEs in the later parts of our Review, we will start our journey through mammalian development with the wide variety of young, often species-specific, TEs and TE-binding KRAB-ZFPs that are expressed prior to implantation.

Pre-implantation development

Zygotic genome activation

At the beginning of mammalian development, the embryo has the potential to form all embryonic and extra-embryonic cell types[65,66]. This state is called totipotency and includes the fertilized egg, the zygote (FIG. 2). Although the impact of TEs on the germ line is discussed in Supplementary Box S1, we focus here on one crucial step in the totipotent phase, zygotic genome activation (ZGA). During ZGA, hundreds of mouse-specific (evolutionarily young) MuERV-L elements are transcribed[24]. The LTRs of several of these MuERV-L elements act as alternative promoters of protein-coding genes[24]. These MuERV-L LTR-derived isoforms include isoforms of developmental regulators and may allow their stage-specific expression[24]. The stage-specific activation of MuERV-L elements partly depends on the eutherian-specific transcription factor DUX[67,68]. Loss of *Dux* in mice reduces MuERV-L and ZGA-specific gene activation and significantly decreases (but does not eliminate) the viability of embryos[69]. It is thus unclear whether DUX-induced MERV-L elements are necessary for pre-implantation development, a question we will return to in the context of the non-LTR retrotransposon L1, which is also highly expressed during ZGA[70]. Whereas MERV-L elements are mouse-specific, DUX has a human orthologue, called DUX4 (REF.[68]). Even though DUX4 differs from mouse DUX in its homeobox domain and DNA-binding motif, DUX4 also plays a role in human ZGA and the activation of a related family of ERVs called *HERV-L* elements, which independently colonized the human genome[68,71]. Divergent DUX-like



Amniotes
Mammals
Ma
420
320
180
Live-born mammals (theria)
160
Coelocanth
Amphibia
Reptiles
Birds
Monotremes
Marsupials
Eutheria
Primates
a Genomes of live-born mammals were colonized by new ERV sequences
L. chalumnae
C. porosus
G. gallus
O. anatinus
M. domestica
M. musculus
M. mulatta
H. sapiens
TE content
2.2 Gb
2.1 Gb
1 Gb
1.8 Gb
3.5 Gb
2.7 Gb
2.6 Gb
3 Gb
Simple repeats/ satellites
DNA transposons
Not annotated as TEs
Retrotransposons
SINE
LINE
LTR/ERV
b Genomes of live-born mammals encode >200 KRAB-ZFPs
KRAB-ZFP count
1,000
500
0
c KRAB-ZFPs can recruit chromatin modifiers
SETDB1
DNMTs
H3K9me3
KAP1
KRAB-ZFPs
5mC
C
TEs
d A subset of mouse and human KRAB-ZFPs are preserved in all live-born mammals
Preservation of mouse KRAB-ZFPs across vertebrate species (ref.[17])
Mouse KRAB-ZFPs
KRAB
Therian-preserved KRAB-ZFPs
Mouse-specific KRAB-ZFPs
e Mouse-specific ERVs such as this ETn element are bound by mouse-specific KRAB-ZFPs
LTR
ETn
LTR
KAP1
H3K9me3
GM13051
ZFP738
ZFP600
REX2
ZFP961
Preservation of human KRAB-ZFPs across vertebrate species (ref.[17])
Human KRAB-ZFPs
KRAB
f Most human KRAB-ZFPs bind to TEs
KRAB-ZFP exo-ChIP–seq
% peaks
100
0
TSS
Simple repeats
TEs
LINE
ERV
SINE
DNA
SVA
KRAB-ZFPs
Orthologues of KRAB-ZFPs (from the indicated species▽) present among 191 vertebrate species (ref.[17])

◀ Fig. 1 | **Evolution of mammalian live birth coincided with novel TE insertions and the co-evolution and conservation of abundant TE-binding KRAB-ZFPs. a** | Composition of transposable elements (TEs) in selected vertebrate genomes[6]. Retrotransposons make up half of mammalian genomes with many long terminal repeat (LTR)-containing endogenous retroviruses (ERVs) in all live-born mammals (theria). By contrast, only 1–2% of mouse and human genomes consists of genes (not shown)[1,3]. **b** | Colonization of therian genomes with many ERVs coincided with the expansion of the family of TE-binding Krüppel-associated box zinc finger proteins (KRAB-ZFPs). KRAB-ZFP counts reflect clusters of KRAB-ZFP orthologues in 191 vertebrate species[17]. Unlike other sequenced vertebrates, not a single live-born mammal has fewer than 200 KRAB-ZFPs. **c** | KRAB-ZFPs bind TEs with arrays of zinc fingers. Many mammalian KRAB-ZFPs use their KRAB domain to recruit the cofactor KRAB-associated protein 1 (KAP1) that recruits chromatin modifiers that can induce heterochromatin through histone modifications and/or 5-methyl-cytosine (5mC). 5mC can also recruit the KRAB-ZFP–KAP1 complex to spread silencing (shown for the individual KRAB-ZFPs ZFP57/ZFP445). **d** | Preservation of mouse and human KRAB-ZFP orthologues across the vertebrate species in part **b**. Eutheria have retained a substantial proportion of therian-specific KRAB-ZFPs, but also contain many species-specific KRAB-ZFPs. This chart is biased by the different evolutionary distances between species in this data set, for example abundant closely-related primate species. Data in parts **b** and **d** are re-analysed from REF.[17] **e** | Occupancy of a mouse-specific early transposon (ETn) by KAP1, trimethylated histone H3 lysine 9 (H3K9me3) and selected mouse-specific KRAB-ZFPs determined by chromatin immunoprecipitation followed by sequencing (ChIP–seq) in mouse embryonic stem cells (ESCs). KRAB-ZFPs bind specifically, and often redundantly across TE sequences. Within ETns, which are non-coding, the motif of the KRAB-ZFP ZFP961 in internal regions resembles a specific primer binding site ($PBS^{Lys1,2}$) present in coding ERVs, such as murine leukaemia virus that is required for priming retroviral reverse transcription and targeted by KAP1-mediated retroviral repression[26,230]. REX2 and ZFP600 or GM13051 also bind to internal regions that in other ERVs often encode packaging signals or highly structured mRNA export signals, respectively[26]. KRAB-ZFPs may thus have evolved to target essential sequences within TEs that cannot easily evade repression by mutation. **f** | Distribution of individual human KRAB-ZFP exo-ChIP–seq peaks in HEK293 cells (that overexpress the factors) among transcription start sites (TSS), or the indicated repetitive sequences. *C. porosus*, *Crocodylus porosus*; DNMTs, DNA methyltransferases; Gb, gigabases; *G. gallus*, *Gallus gallus*; *H. sapiens*, *Homo sapiens*; *L. chalumnae*, *Latimeria chalumnae*; *M. domestica*, *Monodelphis domestica*; *M. mulatta*, *Macaca mulatta*; *M. musculus*, *Mus musculus*; Ma, million years ago; *O. anatinus*, *Ornithorhynchus anatinus*; SETDB1, SET (Su(var)3–9, Enhancer-of-zeste and Trithorax) domain bifurcated histone lysine methyltransferase 1; SINE/LINE, short/long interspersed element; SVA, SINE variable-number tandem-repeat Alu. Part **e** is adapted from REF.[26], CC BY 4.0 (https://creativecommons.org/licenses/by/4.0/). Part **f** is adapted from REF.[17], Springer Nature Limited.

family members, early zygotic genes and ERVs in different mammals are thus an astonishing example of convergent evolution (FIG. 2a).

Is TE expression prior to implantation essential? Although TEs are abundantly expressed in pre-implantation development (particularly during ZGA), the impact of TEs on organismal fitness is an important topic of debate[72]. This debate centres on whether TE activities are essential for the host, and if so, whether it is the transcription of TEs that is necessary or the TE-derived transcripts (which can be coding and/or non-coding). Two recent provocative studies focusing on L1 elements have proposed an important role for these elements in pre-implantation embryos, particularly during ZGA, albeit via distinct mechanisms. The first study demonstrated (using transcription activator-like effector (TALE)-based transcriptional silencing in zygotes or prolonged activation at the two-cell stage) that altered L1 transcription reduced developmental progression to the blastocyst stage by 50%[70]. Blocking reverse transcription or introducing full-length L1 transcripts in the cytoplasm did not recapitulate the phenotype, suggesting that cytoplasmic L1 transcripts and retrotransposition are dispensable for development. Instead, the authors propose that L1-mediated genome-wide chromatin accessibility changes may contribute to the phenotype, yet inducing chromatin decondensation alone only delayed, but did not arrest, developmental progression[70]. This study thus left open whether the partial arrest may be due to nuclear L1 transcripts themselves[70]. The second study used biochemical approaches and antisense oligonucleotides to reduce L1 RNA levels[73]. The authors propose that L1 transcripts themselves are critical for pre-implantation development by directly and positively regulating rRNA genes and negatively regulating the transcription factor *Dux* through interactions with KAP1 and nucleolin[73]. Although compelling, both studies highlight the difficulty of performing and interpreting loss-of-function studies of TEs: the approaches used (whether TALE-based or antisense oligonucleotides) are almost certain to have off-target effects and cannot be easily rescued[74]. These studies therefore do not yet provide the genetic evidence typically required of single-copy genes and long non-coding RNAs (lncRNAs), leaving open the possibility that L1 elements are expressed in this developmental window predominantly for selfish purposes[75]. This latter interpretation is supported by the finding of high expression of species-restricted TEs during ZGA across a wide array of mammals[72].

Long interspersed elements
(LINEs). Autonomous non-LTR retrotransposons that comprise the largest transposable element (TE) class in human and mice (~22% of total genomic content).

Short interspersed elements
(SINEs). Non-autonomous non-LTR retrotransposons that rely on the long interspersed element (LINE) machinery to mobilize and are abundant in mammalian genomes (~7.5% mouse, ~13.5% human).

Extra-embryonic tissue
Shortly after ZGA, totipotency ends with the formation of a cell type that is only found in mammals, the extra-embryonic trophectoderm. The trophectoderm forms by positional cues through separation from the inner-cell mass[76,77]. In mice, its specification is completed by the blastocyst stage[78]. The trophectoderm implants the embryo and later contributes to the placenta (BOX 3; FIG. 2). In mouse trophoblast stem cells (TSCs), an in vitro model of the trophectoderm, 10% of putative enhancers are species-specific[7]. Eighty per cent of these species-specific putative TSC enhancers are derived from species-specific TEs, especially LTRs[7]. These LTR-derived putative enhancers are inactive in mouse embryonic stem cells (ESCs) and adult tissues, and less accessible in in vitro differentiated trophoblast giant cells[7,79]. The activity of these LTRs is therefore not only species-specific but also cell type and stage-specific.

Mouse LTRs of the RLTR13 subfamily provide a striking example of co-option in extra-embryonic cells: containing binding motifs for essential trophectoderm regulators, CDX2, ELF5 and EOMES, RLTR13D5 elements provide a third (>300) of the putative enhancers co-bound by the three factors[7,38] (FIG. 2b). Two of these elements were proven to act as bona fide trophectoderm enhancers as their deletion significantly reduced expression of nearby genes in mouse TSCs[38]. Although this experiment demonstrated the functional importance of TEs as enhancers, and thus their co-option for host function, it does not allow conclusions to be drawn about whether co-opted LTR-derived enhancers in extra-embryonic cells are essential for development because both nearby genes in this experiment

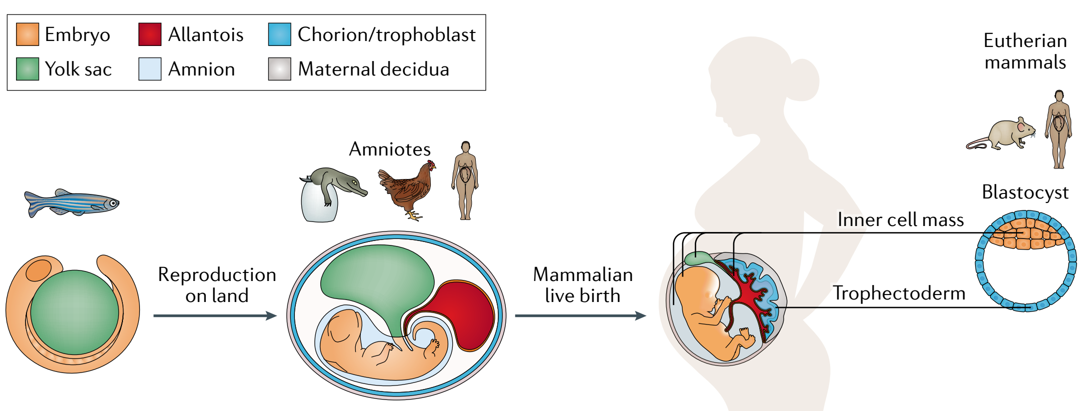
Box 3 | The evolution of mammalian live birth

What extra-embryonic tissues do mammals have?
All amniotes (birds, reptiles and mammals) have four extra-embryonic tissues: the amnion, the allantois, the chorion and the evolutionary ancient yolk sac[20,265,266]. Extra-embryonic tissues retain water (amnion), store metabolic waste (allantois) and exchange gases with the environment (chorion), while also feeding the embryo (yolk sac). These functions are essential to develop on land, which distinguishes amniote development from that of fish and amphibia. In mammals, the chorion develops from the mammalian-specific trophectoderm, which forms the outer layer of the blastocyst in mice and gives rise to the bulk of placental cell types. When trophectoderm comes into direct contact with maternal tissue, which only happens in theria, it attaches the embryo to the uterus, leading to an inflammatory response that induces either birth (marsupials) or implantation and sustained pregnancy (eutheria)[151,195,267].

What exactly is the mammalian placenta made of?
It depends. Most placentae in monotremes and marsupials derive from trophectoderm and yolk sac. Whereas eutherian embryos initially also depend on a (sometimes transient) yolk sac placenta, their mature placenta (which forms halfway through mouse development) has three parts: trophectoderm-derived cell types, maternal tissue and the allantois[268] (see the figure; more detailed schematics and photographs can be found in REFS[20,269,270]). The allantois leaves a visible mark on all of us: our belly buttons, which are left behind by the umbilical cord that develops from the allantois. As both egg-laying and live-born mammals have placentae, a placenta alone is not sufficient for mammals to be live-born — maternal adaptations are also required.

What is the function of the placenta?
The mammalian placenta regulates embryonic growth, organ development and the timing of birth, amongst others, by secreting hormones[151]. However, early in development the developing placenta and other extra-embryonic tissues have an additional role that is often overlooked: embryonic patterning. When the placenta precursor or yolk sac precursor tissue are non-functional or removed, mouse embryos fail to undergo gastrulation, the process that specifies embryonic cell types[148,271,272]. In consequence, embryos cannot develop germ cells, a heart, a gut tube or a brain[115,273]. Without extra-embryonic tissues that form the placenta, mammals thus cannot develop and grow and would not have evolved.

Why are eutherian pregnancies so long?
As maternal and fetal tissues in the eutherian placenta are tightly connected, trophectoderm-derived trophoblast giant cells in mice or extravillous trophoblast cells in humans can control blood flow from the mother by invading her blood vessels. This invasion made nutrient exchange between mother and baby efficient and long-lasting, and explains why marsupial pregnancies are short, but eutheria can develop for much longer (in extreme cases for almost 2 years, for example in elephants and whales)[270]. For the mother, invasive placentation is dangerous. She can only survive pregnancy because mammalian-specific blood cells, the platelets, efficiently repair her blood vessels[274]. These platelets may be one reason why invasive placentation is restricted to mammals, unlike other forms of placentation[206,275]. Prolonged gestation times resulting from invasive placentation allowed greater maturity at birth.



SVA
(SINE variable-number tandem-repeat Alu). Hominid-specific chimeric non-autonomous (long interspersed element 1 (L1)-dependent) retroelements with variable copy number (~2,700 copies in humans, with some of them active) that are a composite of short interspersed element (SINE)-R, variable-number tandem-repeat and Alu elements.

Implantation
The process by which eutherian embryos attach to the uterine wall to establish a sustained maternal–fetal interface that maintains pregnancy. Implantation induces an inflammatory response and can be followed by invasion.

Gastrulation
The embryonic process, lasting for ~72 h from days 6 to 8 in mouse development, that specifies somatic cell fates and a subset of extra-embryonic cell types as well as the germ line, after which twins can no longer form.

Co-option
A process by which natural selection repurposes transposable elements (TEs) for host functions.

Long interspersed element 1
(L1; also known as LINE-1). An active mammalian LINE family in mouse and human (representing 17–20% of their genomes).

DNA methylation
Covalent chemical modification in the form of methylation of cytosines at the fifth position (5-methyl-cytosine (5mC)) that in mammals occurs genome wide predominantly at CpG dinucleotides.

DUF3669
A domain of unknown function found in a few ancient Krüppel-associated box zinc finger proteins (KRAB-ZFPs) that may aid in KRAB-ZFP oligomerization.

SCAN
A *gag* capsid protein-derived domain found in selected ancient Krüppel-associated box zinc finger proteins (KRAB-ZFPs) that mediates protein interactions.

were non-essential genes. Nevertheless, these TEs were co-opted for extra-embryonic gene expression in a species-specific manner, a finding later also shown in other mammals[80].

In human term placenta, for example, putative enhancers were enriched for human ERVs more so than in other tissues[80]. Intriguingly, ERV-derived putative enhancers in human placenta were also the least conserved when compared with putative enhancers in 13 other human tissues[80]. The known high level of morphological variation in placental development across mammals may thus be a result of species-specific ERV colonizations and their activities[7,81,82].

At the blastocyst stage, eutherian embryos specify a second extra-embryonic tissue, the extra-embryonic endoderm. The extra-embryonic endoderm is crucial for the formation of the mammalian yolk sacs and also contributes to embryonic endodermal organs such as the intestine, liver and thyroid, but has yet to be studied with respect to TE functions[83] (FIG. 2).

Pluripotency

The specification of the first extra-embryonic tissues prior to implantation leave the embryonic lineage able to form all embryonic cell types including the mammalian germ line. This ability, called pluripotency,

Zinc fingerprint amino acids
Amino acids at positions −1, +2, +3, and +6 of the C2H2 α-helix that are responsible for direct interactions with specific DNA bases, thus determining the specificity of the zinc fingers.

Zygotic genome activation
(ZGA). The process during which transcription from maternal and paternal genomes in the embryo takes over from maternal control. It occurs at the two-cell or four-cell stage in mice and humans, respectively.

MuERV-L
A murine-specific family of long terminal repeat (LTR) retrotransposons that are highly transcribed at zygotic genome activation (ZGA), present in ~1,500 copies. These elements are related to the endogenous retrovirus L (ERV-L) family of retrotransposons found throughout eutherian mammals that include human ERV-L (HERV-L) elements found in ~2,000 copies.

Long non-coding RNAs
(lncRNAs). RNA molecules >200 nucleotides long that do not encode proteins.

Trophectoderm
A mammalian-specific extra-embryonic cell lineage that gives rise to invasive trophoblast giant cells that mediate implantation in eutheria and to the chorion that contributes to all mammalian placentae.

Blastocyst
A thin-walled hollow embryonic structure containing a eutherian-specific inner cell mass, the future embryo and — dependent on its maturation — one or two extra-embryonic cell lineages.

Trophoblast stem cells
(TSCs). Undifferentiated extra-embryonic stem cells isolated from the trophectoderm in the blastocyst or TSCs shortly after implantation that can functionally contribute to placental development when reintroduced into blastocysts.

Enhancers
Non-coding short (typically 100–1,000 bp in length) DNA sequences that act to drive transcription independent of their relative distance, location or orientation to their cognate promoter. Putative enhancers bear histone modifications such as acetylated histone H3 lysine 27 (H3K27ac) and methylated H3 lysine 4 (H3K4me1).

depends on essential pluripotency factors including OCT4 (REFS[84,85]).

ERVs provide species-specific enhancer activity in pluripotent cells. Mouse and human ESCs share only 5% of OCT4 binding sites[86]. ERV-K elements provide up to a quarter of OCT4 binding sites in mouse and human ESCs, suggesting that these ERVs contribute to species-specific OCT4 binding[86,87]. Although many transcription factors derive their species-specific binding sites from TEs, considerable experimental evidence supports that OCT4-bound ERV-K elements also act as species-specific enhancers[30,88,89]. In human pluripotent embryonic carcinoma cells (EC cells), CRISPR activation/interference screens targeting a subgroup of OCT4 motif-containing evolutionarily young ERV-K elements called LTR5HS (~700 elements) revealed that 71% of the nearby genes were differentially expressed[27,39] (FIG. 2c). Furthermore, deletion of six individual LTR5HS elements reduced nearby gene expression, as did deletion of two independently identified pluripotency factor-bound RLTR13D6 elements in mouse ESCs that frequently interact with nearby promoters[38,39]. By contrast, a CRISPR interference screen in mouse ESCs that simultaneously targeted >400 pluripotency factor-bound RLTR13D6 elements minimally influenced gene expression[38]. The results of these experiments provide evidence that ERV-derived regulatory sequences can, but do not always, regulate host genes, and that such regulation is likely to be dependent upon additional locus-specific interactions.

Human ERVs are part of pluripotency-associated lncRNAs. Human ESCs express several hundred primate-specific human endogenous retrovirus-H (HERV-H) elements, ~10% of which produce chimeric transcripts with host transcripts as lncRNAs[90–93]. The LTR portion of these HERV-H elements, LTR7, has promoter activity that is specific to human pluripotent cells and is often bound by OCT4 (REFS[91–93]). In human ESCs, knock down of HERV-H induces differentiation and affects self-renewal, a phenotype similarly seen for individual HERV-H-containing lncRNAs, most notably the highly expressed ESRG[92,93]. Yet the evidence supporting an important role for HERV-H-containing lncRNAs in pluripotency is based on lncRNA knock-down studies, which are less effective for depleting nuclear lncRNAs and can be biased by off-target effects[74,92,93]. Furthermore, these phenotypes were not shown to be restored by expressing any HERV-H transcripts from a transgene resistant to the RNA interference molecule, leaving open the possibility that HERV-H elements are not essential or merely modulatory[75]. Supporting the need for careful genetic studies, a recent knockout of the aforementioned ESRG lncRNA, unlike its knock down, was dispensable for the identity of pluripotent cells[94]. Thus, the role of HERV-H-derived lncRNAs in mammalian development has yet to be fully elucidated.

Possible reasons underlying high TE expression prior to implantation. Although most TEs become progressively silenced after implantation (see next section), pre-implantation embryos, extra-embryonic cells and the germ line (see Supplementary Box S1) abundantly express TEs. These developmental contexts share two features: they allow germ-line transmission and undergo epigenetic reprogramming. Mammalian totipotent and pluripotent cells can give rise to germ cells and hence, like germ cells, allow direct germ-line transmission (see Supplementary Box S1). Whereas extra-embryonic cells cannot directly contribute to the germ line, trophoblast cells can be infected both by exogenous viruses such as HIV-1 as well as by ERVs[95,96]. For example, enJSRV, which is expressed in the uterine epithelium of female sheep, can infect the trophectoderm of transplanted cattle blastocysts that lack this ERV[95]. Moreover, via the placenta, fetal cells can colonize the mother and vice versa[97]. The placenta thus allows indirect germ-line transmission. Although speculative, 'selfish' ERVs could thus propagate from the mother to the fetal germ line; by infecting the mother or her oocytes, fetal trophoblast-expressed ERVs could even enter the germ line of littermates or future offspring (discussed in more detail in REF.[82]). From a selfish perspective, TEs evolving to be expressed in early or extra-embryonic development thus maximize germ-line propagation[98]. In addition, active and passive DNA demethylation shortly after fertilization and during germ-line development could permit TE expression[99]. The 5mC levels in extra-embryonic tissues of the developing and mature placenta are also globally lower compared with somatic tissues[99]. Although the repression of TEs such as intracisternal A-type particles (IAPs) after implantation clearly depends on 5mC, a causal relationship between TE expression and DNA demethylation prior to implantation has not been formally established[52].

Stage-specific KRAB-ZFP repression may enable TE co-option. Despite the widespread expression of retrotransposons, species-specific KRAB-ZFPs and KAP1 are also highly expressed in mouse and human pluripotent cells[17,26,46,62,63,100]. In mouse or human ESCs, the KRAB-ZFP–KAP1 system broadly represses TEs as many TE-adjacent genes are derepressed when *Trim28* is removed[46,101]. However, the rapid death of KAP1-depleted ESCs precludes further functional studies[46,101]. To overcome this limitation, recent experiments systematically deleted entire clusters of KRAB-ZFPs in mouse ESCs and mice[26]. These experiments revealed widespread derepression of specific TE families and concomitant reactivation of putative TE-derived enhancers, although development remained undisturbed[26]. For example, loss of a KRAB-ZFP cluster on chromosome 4 leads to KAP1/H3K9me3 loss and H3K4me3/acetylated H3 lysine 27 (H3K27ac) gain at early transposon (ETn) elements, suggesting these ETn elements may be dormant enhancers. Indeed, loss of an individual KRAB-ZFP-bound ETn element derepresses a nearby gene[26]. This experiment indicated that this KRAB-ZFP suppresses a putative ETn-derived enhancer and may regulate gene expression rather than merely repress a TE[26]. In human ESCs, this function in gene regulation convergently evolved and may be developmentally relevant: in this case, KRAB-ZFPs repress primate-specific short interspersed nuclear element (SINE) variable-number tandem-repeat Alu (SVA) elements in pluripotent stem

Embryonic stem cells
(ESCs). Pluripotent stem cells derived from the inner cell mass of the blastocyst that can functionally contribute to the germ line and somatic tissues when reintroduced into blastocysts.

Trophoblast giant cells
Heterogeneous, large, polyploid trophectoderm-derived cells that mediate implantation and decidualization but also line maternal blood vessels and produce hormones in the placenta.

RLTR13
Mouse-specific endogenous retroviruses (ERVs) of the ERV-K family (K standing for the lysine tRNA primer used during reverse transcription) with many subfamilies, for example RLTR13D5 (~685 copies) or RLTR13D6 (~790 copies).

Amnion
Extra-embryonic tissue in amniotes that fills with fluid to form a sac that protects the embryo and provides a watery environment for developing on land.

Allantois
An extra-embryonic mesoderm-derived vascularized sac-like structure that allows gas exchange and disposes of liquid waste. It forms part of the eutherian placenta and becomes the umbilical cord.

Chorion
An extra-embryonic tissue that exchanges gases with the environment and originated in amniotes.

Yolk sac
The oldest extra-embryonic tissue in amniotes that originally absorbed nutrients deposited in yolk but has — despite the lack of yolk in eutherian mammals — remained essential for embryonic development, for example for embryonic patterning and blood development.

Extravillous trophoblast cells
Invasive, differentiated human trophoblast cells (equivalent to mouse trophoblast giant cells) that invade deeply into the uterine stroma, where they line and remodel maternal spiral arteries to supply the placenta with maternal blood.

cells that later act as neuronal enhancers[102] (FIG. 2d). These experiments suggest that the specific repression of TEs by KRAB-ZFPs in early development, when their activity may be detrimental, may allow TEs to be co-opted as tissue or stage-specific enhancers later in development.

Origin of an essential pluripotency gene and mammalian-specific genes from transposases. Mammalian genomes encode a family of ancient vertebrate THAP domain proteins, which are putative transcription factors that derive their DNA-binding THAP domain from the DNA-binding domain of P element-like transposases[103]. One of these, THAP11, is essential for mouse ESC maintenance and progression through the blastocyst stage in vivo[104–108] (FIG. 2e). *Thap11* overexpression maintains pluripotency even in the absence of OCT4, suggesting a critical function in ESC viability[109,110]. This function may be evolutionary conserved, as THAP11 co-binds promoters of cell cycle and metabolism genes with ZNF143 and the transcription factor HCF1 in human ESCs[105,111,112]. In humans, cell proliferation depends on the HCF1 interaction via the dHsY motif of THAP11 that is conserved in other THAP proteins[113]. Although THAP11 is an essential transposase-derived gene in mice, it is unclear what aspect of this function is a unique innovation in mammals as opposed to its ancestral function in other vertebrates. In contrast to *Thap11*, which is a mono-exonic gene, other Thap genes are multi-exonic and express their protein products through splicing of TE domains to host protein domains. The recent discovery of ~100 such host–transposase fusions (HTFs) in tetrapods suggests that many genes evolved this way[114]. These HTFs underwent purifying selection and are expressed in human tissues, suggesting they may have important functions[103,114]. Although all vertebrate lineages have HTFs, there are apparent bursts throughout mammalian evolution (that is, therian and eutherian ancestors), which could suggest they play a role in mammalian development (FIG. 1e). It will be exciting to discover how TE-derived DNA-binding domains of transposases have impacted the evolution of mammalian developmental transcription factors.

Taken together, TEs affect gene expression during pre-implantation development in three major ways: first, as transcripts; second, as regulatory elements; and third, as building blocks for genic isoforms and new genes. Some of the TE-derived genes such as the transcription factor THAP11 are clearly essential, whereas the developmental importance of the large number of regulatory elements and transcripts is unknown. These TE transcripts and regulatory activities occur, however, at a time in development when the mammalian embryo is epigenetically permissive, so their presence is hardly surprising. Interestingly, some of the observed co-option events may have been facilitated by KRAB-ZFPs. These KRAB-ZFPs initially silenced newly integrated TEs until both KRAB-ZFP and targeted TE became integrated in tissue or stage-specific gene regulatory networks, further illustrating the inseparable co-evolution of TEs and defence systems.

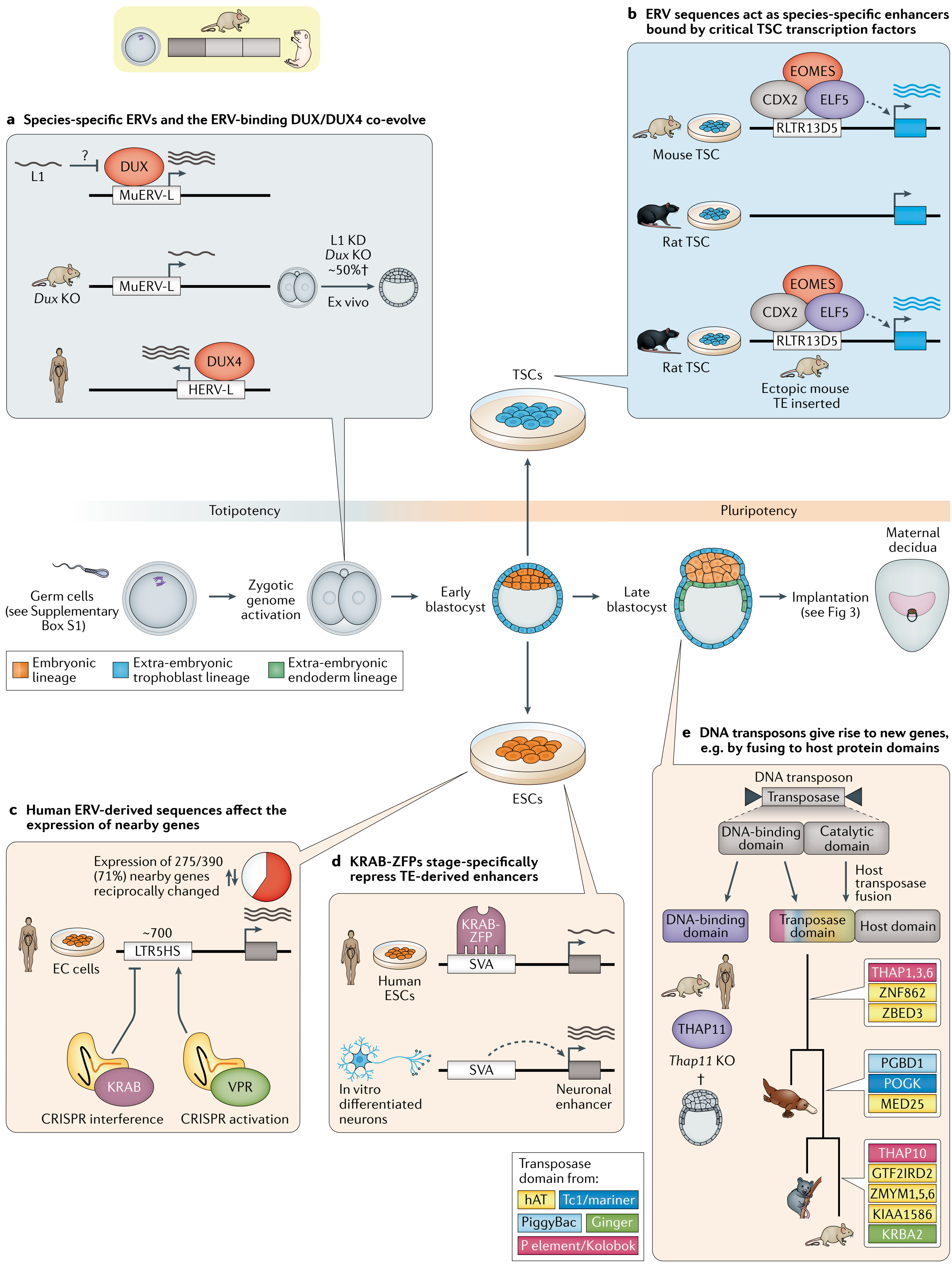
Embryonic–extra-embryonic interface

Gastrulation

After mouse and human implantation, the blastocyst develops into a fetus with organs and a functional placenta that sustains it until birth. Before embryonic organs or the placenta can form, the embryo lays out its basic body plan during the process of gastrulation, which specifies embryonic cell types between days 6 and 8 of mouse development and from weeks 2 to 3 of human development[115,116] (FIG. 3).

Tissue-specific restriction of TE expression prior to gastrulation. Expression of LINEs, SINEs and some LTR-containing retrotransposons peaks before or around implantation, and ceases in the embryo prior to gastrulation when 5mC covers these TE families specifically

Fig. 2 | **Species-specific TEs and KRAB-ZFPs promote the rapid evolution of mammalian pre-implantation development. a** | In an astonishing example of convergent evolution, mouse MuERV-L and human HERV-L elements are bound and induced by the transcription factor DUX/DUX4 during zygotic genome activation (ZGA). In mice, DUX may be repressed by long interspersed nuclear element 1 (L1) transcripts via interactions with the Krüppel-associated box zinc finger protein (KRAB-ZFP) cofactor KRAB-associated protein 1 (KAP1) and the conserved ribosomal maturation protein nucleolin[73]. L1 knock down (KD) or *Dux* knockout (KO) arrests (†) many mouse embryos ex vivo (that is, outside the context of the mother). **b** | RLTR13D5 endogenous retroviruses (ERVs) generated novel binding sites for conserved trophectoderm regulators that can act as species-specific enhancers in mouse trophoblast stem cells (TSCs). Rats lack these RLTR13D5-derived enhancers, but can drive mouse-specific expression when the mouse RLTR13D5-derived enhancer is ectopically introduced. **c** | Large-scale CRISPR activation or CRISPR interference screens have allowed the manipulation of whole transposable element (TE) subfamilies. Such an experiment for targeting ~700 human LTR5HS elements in pluripotent human cells demonstrated reciprocal changes in the expression of 275 of 390 nearby genes in a CRISPR interference (repressive dCas9–KRAB) and CRISPR activation (transactivating dCas9–VPR) screen (pie chart). **d** | Species-specific human KRAB-ZFPs bind to species-specific TEs to repress nearby genes, which were shown to be poised for later activation during in vitro differentiation into neurons. This experiment suggests that KRAB-ZFPs allow the co-option of TEs for stage-specific developmental gene regulation. **e** | DNA-binding domains of the transposases encoded in DNA transposons gave rise to TE-derived proteins. These include, for example, the transcription factor THAP11, which is derived from a P element-like transposase DNA-binding domain and is essential in mouse embryonic stem cells (ESCs). DNA transposons can also give rise to host–transposase fusions (HTFs) in which one of the transposase domains, most often a DNA-binding or catalytic domain, is fused to a host protein domain. Such HTFs of as yet unknown function arose at critical breakpoints in mammalian evolution. The host domains for these mammalian HTFs can be found in Table S3 of REF.[15]. See Supplementary Box S1 for roles of TEs and TE repressors in the mammalian germ line. EC cells, embryonic carcinoma cells; hAT, hobo Activator Tam3; SVA, short interspersed nuclear element (SINE) variable-number tandem-repeat Alu.



a Species-specific ERVs and the ERV-binding DUX/DUX4 co-evolve
L1
?
DUX
MuERV-L
Dux KO
MuERV-L
L1 KD
Dux KO
~50%†
Ex vivo
DUX4
HERV-L
b ERV sequences act as species-specific enhancers bound by critical TSC transcription factors
EOMES
CDX2
ELF5
RLTR13D5
Mouse TSC
Rat TSC
EOMES
CDX2
ELF5
RLTR13D5
Rat TSC
Ectopic mouse TE inserted
TSCs
Totipotency
Pluripotency
Germ cells (see Supplementary Box S1)
Zygotic genome activation
Early blastocyst
Late blastocyst
Implantation (see Fig 3)
Maternal decidua
Embryonic lineage
Extra-embryonic trophoblast lineage
Extra-embryonic endoderm lineage
ESCs
c Human ERV-derived sequences affect the expression of nearby genes
Expression of 275/390 (71%) nearby genes reciprocally changed
EC cells
~700
LTR5HS
KRAB
VPR
CRISPR interference
CRISPR activation
d KRAB-ZFPs stage-specifically repress TE-derived enhancers
Human ESCs
KRAB-ZFP
SVA
In vitro differentiated neurons
SVA
Neuronal enhancer
e DNA transposons give rise to new genes, e.g. by fusing to host protein domains
DNA transposon
Transposase
DNA-binding domain
Catalytic domain
Host transposase fusion
DNA-binding domain
Tranposase domain
Host domain
THAP11
Thap11 KO
†
THAP1,3,6
ZNF862
ZBED3
PGBD1
POGK
MED25
THAP10
GTF2IRD2
ZMYM1,5,6
KIAA1586
KRBA2
Transposase domain from:
hAT
Tc1/mariner
PiggyBac
Ginger
P element/Kolobok

Extra-embryonic endoderm
The precursor tissue of parts of the mammalian yolk sacs and substantial parts of the developing mammalian gut tube and associated endodermal organs. Extra-embryonic endoderm is also important for embryonic patterning and the specification of embryonic cell types, for example the future brain.

Pluripotency
The ability to functionally contribute to all somatic cells and the germ line of an organism that is maintained after implantation until specific cell types and the germ line are specified through gastrulation.

ERV-K
A subfamily of endogenous retroviruses (ERVs) that uses a lysine (K) tRNA that binds to the viral primer binding site to prime reverse transcription. ERV-K elements in mice contain many mobile elements (for example, intracisternal A-type particle (IAP), early transposon (ETn)).

Embryonic carcinoma cells
(EC cells). Malignant pluripotent cells derived from germ cell tumours called teratocarcinomas. Their culture preceded and paved the way for establishing embryonic stem cells (ESCs).

CRISPR activation/interference
Sequence-specific activation or repression of gene expression on the transcriptional level by exploiting the bacterial genetic immune system clustered regularly interspaced short palindromic repeats (CRISPR) pathway with nuclease-dead Cas9 (dCas9) fused to an activation or repression domain.

LTR5HS
Long terminal repeat (LTR) of one of the youngest families of human endogenous retrovirus-K (HERV-K) elements present in ~700 copies in the human genome.

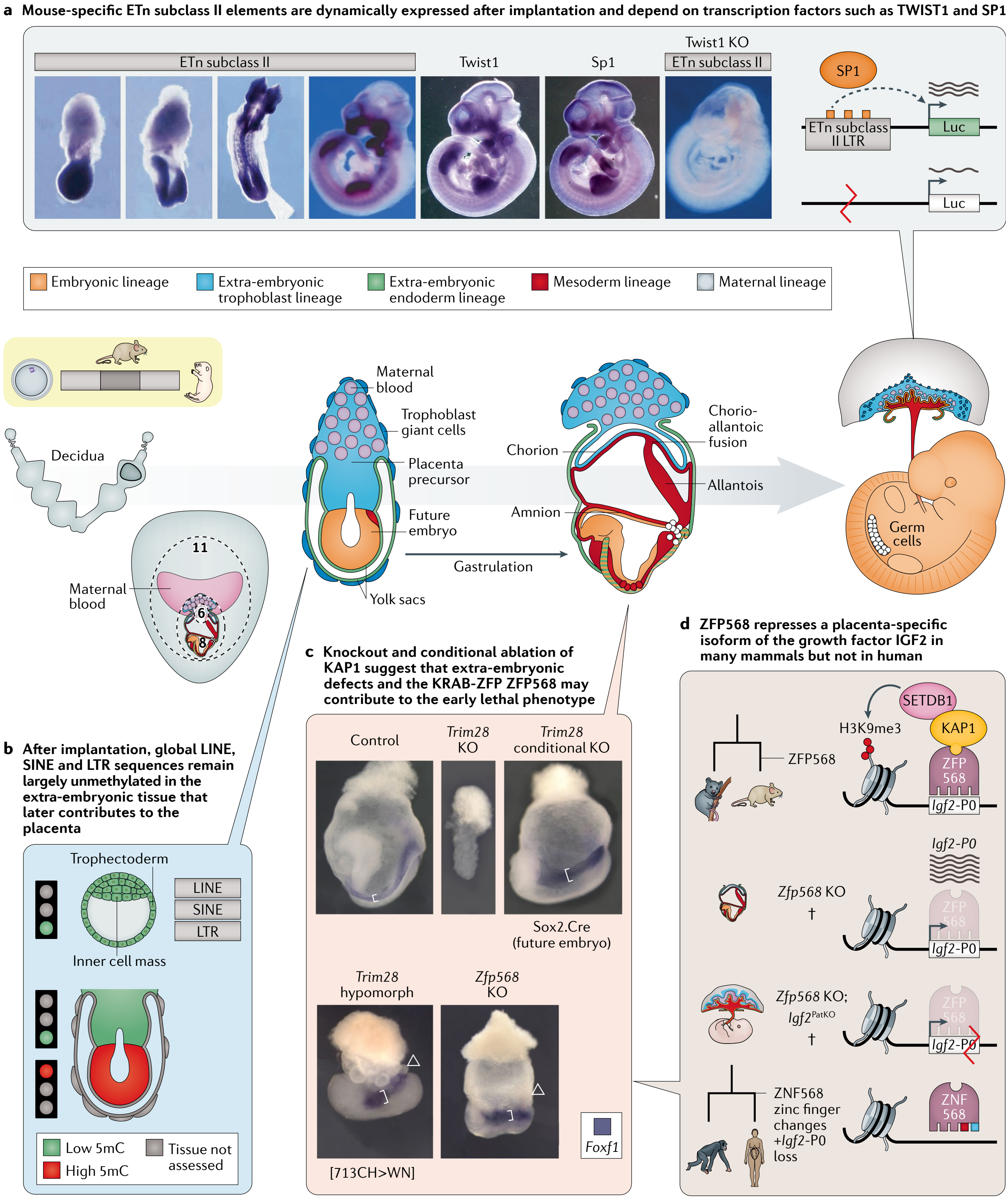
Epigenetic reprogramming
Erasure and remodelling of epigenetic marks, such as DNA methylation and histone modifications, particularly during early development and in the germ line.

in the embryonic lineage[52,99,117–120]. However, LINE, SINE and LTR sequences remain unmethylated in the extra-embryonic tissue that later contributes to the placenta[99,121]. Selected ERVs, for example mouse-specific ETn subclass II elements, remain dynamically expressed in both embryonic and extra-embryonic tissues throughout gastrulation[122,123]. In the embryo, ETn subclass II elements can be detected at least until mid-gestation, when their expression was shown to depend on conserved developmental transcription factors[122,123] (FIG. 3a). Although these ETn subclass II elements may well have a function for the host, they could also simply exploit host transcription factors (here TWIST1 and SP1/3) to be expressed in the early embryo and placenta to propagate (see discussion above). In such a model, their expression in the mid-gestation embryo would just be a by-product, because these transcription factors also happen to be expressed at later developmental stages.

Gastrulation may depend on TE-controlling mechanisms. The technical difficulties associated with genetically ablating abundant, repetitive TE families make it challenging to determine whether TE expression is essential for gastrulation. Nonetheless, two lines of evidence suggest that the silencing of TEs may be essential for development. First, the forced expression of codon-optimized L1 elements during development using an inducible transgene led to a dose-dependent increase in retrotransposition and embryonic lethality, suggesting that a high retrotransposon load can be lethal, although the timing of developmental arrest was not assessed (similarly, ectopic L1 expression reduced viability in cancer cells and pre-implantation mouse embryos)[70,124,125]. Second, loss of several TE regulators leads to lethal gastrulation defects. For example, *Trim28* (encoding KAP1) or *Setdb1* deletion, which causes global ERV reactivation, causes lethality prior to gastrulation[46,49,126,127]. Many other KAP1-interacting chromatin regulators involved in TE regulation also have essential roles during gastrulation, including the three DNA methyltransferases (DNMT1, DNMT3A and DNMT3B), the methylcytosine dioxygenases (TET1, TET2 and TET3) that antagonize DNA methyltransferases as well as components of the HUSH complex (including TASOR and PERIPHILIN1) that silence unintegrated retroviral DNA and young L1 elements[52,128–132] (see Supplementary Table S1). Although these phenotypes could reflect an essential role of TE repression for gastrulation, the normal development of mice that have lost dozens of KRAB-ZFPs and derepress TEs from multiple families suggests that derepression of young TEs can be compatible with life[26]. Detangling TE repression-dependent and repression-independent roles of all KRAB-ZFPs and their co-repressors will be an important avenue for future research.

Ectopic TE expression may also influence extra-embryonic development. The tissue-specific gain in 5mC at repetitive sequences and genome wide shortly after implantation gradually establishes an epigenetic barrier against the expression of extra-embryonic transcripts in the developing embryo that is partially lost when KRAB-ZFP co-repressors are disabled[99,121,133] (FIG. 3b). Without KAP1 or DNA methyltransferases, genes normally confined to extra-embryonic tissues become ectopically expressed[101,134–136]. In line with an important role of these epigenetic pathways during extra-embryonic development, placental defects are common when epigenetic reprogramming is incomplete (for example, during somatic cell nuclear transfer)[136–139]. In addition, many KRAB-ZFP co-repressors and individual KRAB-ZFPs are critical for extra-embryonic development[101,134–136] (see Supplementary Table S1). For mutants of either KAP1 or SETDB1, extra-embryonic defects contribute substantially to the early phenotypes: when both extra-embryonic

Fig. 3 | **Factors that regulate TEs adopted essential roles during early post-implantation development. a** | Mouse-specific early transposons (ETns) of subclass II are tissue and stage-specifically expressed after implantation. These endogenous retroviruses (ERVs) are regulated by the conserved developmental transcription factors TWIST1 and SP1/3 (REFS[23,123,231,232]). Expression patterns from REFS[23,231,233]. These transcription factors mediate embryonic and placental development in mice but are, unlike ETns, deeply conserved, which questions critical roles for ETn elements for their conserved functions[232,234,235]. **b** | Whereas retrotransposons (LINEs, SINEs and LTRs) in the embryonic lineage become gradually repressed after implantation, they remain expressed in the placenta precursor tissue that is mirrored in differential 5-methyl-cytosine (5mC) patterns[99,121,236–238] (at both stages, embryonic and extra-embryonic tissues differ transcriptionally). It is noteworthy that distinct cell types generated through gastrulation differ in their 5mC patterns, with brain precursors being less methylated and developing germ cells secondarily losing most 5mC during a second wave of epigenetic reprogrammng[238]. **c** | Embryonic phenotypes of selected knockout, conditional and hypomorph *Trim28* mutants in comparison with *Zfp568* knockout (KO) and control embryos at day 8 of mouse development. Blue staining depicts *Foxf1* transcripts, which mark lateral plate mesoderm (square bracket) that forms during gastrulation. Triangle indicates a blistered yolk sac. Mouse embryos can initiate gastrulation when *Trim28* is conditionally ablated specifically in the embryonic cell lineage, whereas loss of *Trim28* in both embryonic and extra-embryonic cell lineages arrests development prior to gastrulation[141,239–241]. KRAB-associated protein 1 (KAP1) is thus involved in reciprocal interactions between embryonic and extra-embryonic tissues that initiate gastrulation. **d** | The Krüppel-associated box zinc finger protein (KRAB-ZFP) ZFP568 that evolved in eutheria was repurposed to repress a placenta-specific isoform of the paternally (Pat) expressed growth factor IGF2 (*Igf2*-P0). *Zfp568* knockout embryos produce abundant IGF2 and arrest (†) due to gastrulation defects (see part **c**). Paternal *Igf2* deletion can rescue these gastrulation defects and embryos arrest only shortly before birth. When humans split from chimpanzees, the zinc fingers of ZNF568 mutated and lost the ability to bind *Igf2*-P0. Whether ZNF568 is essential in human is not known. H3K9me3, trimethylated histone H3 lysine 9; LTR, long terminal repeat; SETDB1, SET (Su(var)3–9, Enhancer-of-zeste and Trithorax) domain bifurcated histone lysine methyltransferase 1; SINE/LINE, short/long interspersed nuclear element. Part **a** is adapted with permission from REF.[23], Elsevier, and REF.[231], Wiley, and Twist1 and Sp1 images are courtesy of Q. Ma. Part **c** is adapted with permission of Company of Biologists, from *Development*, Shibata, M., Blauvelt, K. E., Liem, K. F. Jr. & Garcia-Garcia, M. J. **138**, 24, 2011; permission conveyed through Copyright Clearance Center, Inc. (REF.[141]).



and embryonic tissues lack KAP1 or SETDB1, embryos show an early patterning defect and cannot initiate gastrulation[140,141]. By contrast, when only the embryonic lineage lacks either KAP1 or SETDB1, embryonic patterning is partially restored and gastrulation can initiate[140,141]. Even though these conditional mutant embryos cannot complete gastrulation successfully, they develop considerably further than the knockout in both embryonic and extra-embryonic tissues[140,141] (FIG. 3c; see Supplementary Table S1). These findings strongly suggest that the

Intracisternal A-type particles
(IAPs). A rodent-specific and still actively retrotransposing class of transposable elements (TEs). They are among the most mutagenic long terminal repeat (LTR) retrotransposons in mammals and are present at ~1,000 full-length copies per haploid genome in mice.

Early transposon
(ETn). A type of non-autonomous mouse-specific long terminal repeat (LTR) retrotransposon that is mobilized by related MusD elements. They are called 'early transposons' because they are undetectable in differentiated cell lines. They still actively retrotranspose.

P element
A eukaryotic DNA transposon discovered as one of the first transposable elements (TEs) in *Drosophila melanogaster*.

Host–transposase fusions
(HTFs). Fusion proteins combining DNA transposase and host-derived domains that often originate via alternative splicing.

Somatic cell nuclear transfer
Transfer of a nucleus of an adult somatic cell to an enucleated egg cell. Pioneered in frogs, nuclear transfer can be used to clone animals as it leads to epigenetic reprogramming towards pluripotency.

Genetic drift
Random sampling from parental genomes in the next generation that generates changes in the frequency of existing genetic traits in a population.

Maternal–fetal interface
Sites where the genetically distinct mother and fetus come into indirect contact. It can refer both to the contact of mother and fetus in the functional placenta and to the implantation site.

Chorioallantoic fusion
Fusion between the allantois and the chorion at day 8–9 of mouse development that subsequently allows the vasculature of the allantois to grow into chorion-derived villi to form the labyrinth layer of the placenta.

extra-embryonic expression of KAP1 and SETDB1 is important for both embryonic and extra-embryonic development.

To what extent do essential KAP1 functions depend on KRAB-ZFPs? There are at least three KRAB-ZFPs — ZFP57, ZFP110 and ZFP568 — that have essential functions during development and utilize KAP1 and SETDB1 as cofactor and co-repressor, respectively (see Supplementary Table S1). All three evolved in the eutherian ancestor. The earliest known lethal phenotype of a KRAB-ZFP is caused by ablation of *Zfp568* (FIG. 3c). In the embryonic lineage, the functions of KAP1 may largely depend on *Zfp568*, as loss of ZFP568 leads to similar phenotypes to the well-characterized gastrulation defects caused by conditional KAP1 deletion specifically in the embryo or hypomorphic KAP1 mutant embryos that may interfere with recruiting SETDB1 or other co-repressors[62,142] (FIG. 3c; see Supplementary Table S1).

ZFP568 regulates a placenta-specific isoform of the growth factor IGF2. The evolution of ZFP568 may represent a case where TEs have driven the evolution of mammalian development indirectly. Starting at implantation, ZFP568 represses a placenta-specific promoter of the insulin growth factor 2 (*Igf2-P0*) that boosts overall placental IGF2 levels by about 10%[143]. ZFP568-dependent gastrulation defects can be rescued almost up to birth by ablating the paternal *Igf2* allele from which imprinted *Igf2* regulates embryonic growth via the placenta[143]. When modelled in mouse ESCs, *Igf2-P0* repression depends on both KAP1 and SETDB1, as the *Igf2-P0* promoter loses H3K9me3 in the absence of *Zfp568* (REF.[143]) (FIG. 3d). Whereas halving or increasing IGF2 levels up to approximately threefold is compatible with development, the complete derepression and resulting several-fold increased *Igf2* expression in the absence of *Zfp568* is toxic[144,145]. Although *Igf2-P0* is too ancient to confidently determine a possible LTR origin, it is plausible that *Igf2-P0* co-option in the eutherian placenta was only possible by simultaneously repressing it in the embryo via ZFP568. More broadly, many evolutionarily old KRAB-ZFPs such as ZFP568 preferentially bind to promoters rather than extant TEs[17]. It is tempting to speculate that these promoters were themselves derived from ancient TEs that have decayed by genetic drift beyond recognition.

ZNF568 may be a decaying KRAB-ZFP in humans. The zinc finger array and KRAB domains of ZFP568 are highly conserved throughout eutheria, as is its binding site at *Igf2-P0*, suggesting that ZFP568-mediated repression of *Igf2-P0* is essential throughout mammals[146] (FIG. 3d). Humans appear to be a notable exception: they have accumulated mutations within the *Igf2-P0* binding site and elsewhere in the broader *Igf2-P0* promoter region that abolish *Igf2-P0* expression in the human placenta[146]. This loss may explain the reduced constraint on human *ZNF568* and its subsequent rapid evolution: human *ZNF568* has three distinct high-frequency alleles with accumulated mutations in the zinc finger array that all disrupt *Igf2-P0* binding in reporter assays, resulting in loss of H3K9me3 at the *Igf2-P0* region in human ESCs[146,147]. It is plausible that human *ZNF568* and its orthologues in mammals may play additional roles later in development that do not require the entire zinc finger array. Indeed, *Zfp568*-mutant mouse embryos with experimentally reduced IGF2 still die prior to birth, and *Zfp568* conditional knockout in the mouse brain leads to reduced brain weight at birth (that normalizes through to adulthood) and altered gene expression in cultured neural stem cells[147]. Consistent with the association between ZFP568 and neural development in mice, human *ZNF568* alleles are associated with the ratio of head circumference to height at birth[143,147]. Human *ZNF568* could thus either have ancient IGF2-independent functions, have adopted new human-specific functions or be on its way to eventual decay and loss from the genome.

Extra-embryonic tissues, including mammalian-specific placenta precursor cells, have long been known to enable the patterning processes that precede and enable gastrulation[115,148–150]. The *Trim28* (encoding KAP1) knockout phenotype is a powerful example that the ability to initiate gastrulation depends on extra-embryonic functions. Moreover, to complete gastrulation, mouse embryos require the ancient KRAB-ZFP ZFP568 in the embryonic lineage, which uses KAP1 as a cofactor. ZFP568 is thus likely to further illustrate the principle introduced in the last section that co-evolving TEs and cognate KRAB-ZFPs can become part of mammalian-specific gene regulatory networks. In this particular case, a TE-derived regulatory element may have become an integral part of the fundamentally mammalian process to tissue-specifically repress IGF2 expression in the embryo but allow for its expression in the placenta.

Maternal–fetal interactions

To allow the embryo to grow and develop further after gastrulation, eutherian mothers provide oxygen and nutrients whereas the embryo returns metabolic waste. This exchange happens in the placenta via indirect contact of embryonic and maternal blood at the maternal–fetal interface (FIG. 4). The maternal–fetal interface defines a functional eutherian placenta, which is established around mid-gestation (~9–10 days) in mice and at the end of the first trimester in humans[151]. A functional placenta can only form after chorioallantoic fusion, which results from successful gastrulation (FIGS 3,4). Whereas both ZFP568 and KAP1 and other KRAB-ZFP-interacting TE regulators are essential for chorioallantoic fusion at days 8–9 of mouse development[141,142] (see Supplementary Table S1), co-opted TE-derived proteins are essential for establishing a functional placenta at mid-gestation.

Establishing a functional placenta

ERV-derived env and gag proteins contribute to the maternal–fetal interface. In addition to regulatory sequences, ERVs also encode proteins. The co-option of several ERV-encoded proteins for mammalian placental functions is one of the most striking examples of TE co-option. In contrast to co-opted regulatory elements, the loss of co-opted TE-derived proteins unambiguously leads to embryonic lethality in mice. At the

Syncytiotrophoblast
Multinucleated layers (two in mice, one in humans) in the placenta that are in indirect contact with maternal blood and act as a transport surface.

env
Retroviral gene encoding a cell surface protein that mediates viral–host cell fusion.

gag
Retroviral genes encoding structural proteins of retroviruses that contribute to capsid formation and are essential for infectivity.

pol
Retroviral genes encoding proteins with enzymatic activities required for viral replication (for example, reverse transcriptase), integration and protein cleavage.

sushi-ichi
Long terminal repeat (LTR) retrotransposons of the Ty3/Gypsy class that encode chromodomain integrases, a primer binding site, two open reading frames for *gag* and *pol* genes and a polypurine tract.

Labyrinth
The innermost layer of the mouse placenta that is the main site of nutrient and gas exchange and consists of the syncytiotrophoblast and the vascular network.

Spongiotrophoblast
The hormone-secreting, trophectoderm-derived middle layer of the mouse placenta sandwiched between the outer secondary trophoblast giant cells and the inner labyrinth layer with poorly understood, yet essential, functions in pregnancy.

Genomic imprinting
The parent of origin-specific expression of genes that is established during germ-line development and depends on differential DNA methylation of so-called imprinting control regions (ICRs). Canonical imprints solely depend on DNA methylation, whereas non-canonical imprints (prevalent in the placenta) initially depend on maternally inherited repressive histone marks that are replaced by DNA methylation after implantation.

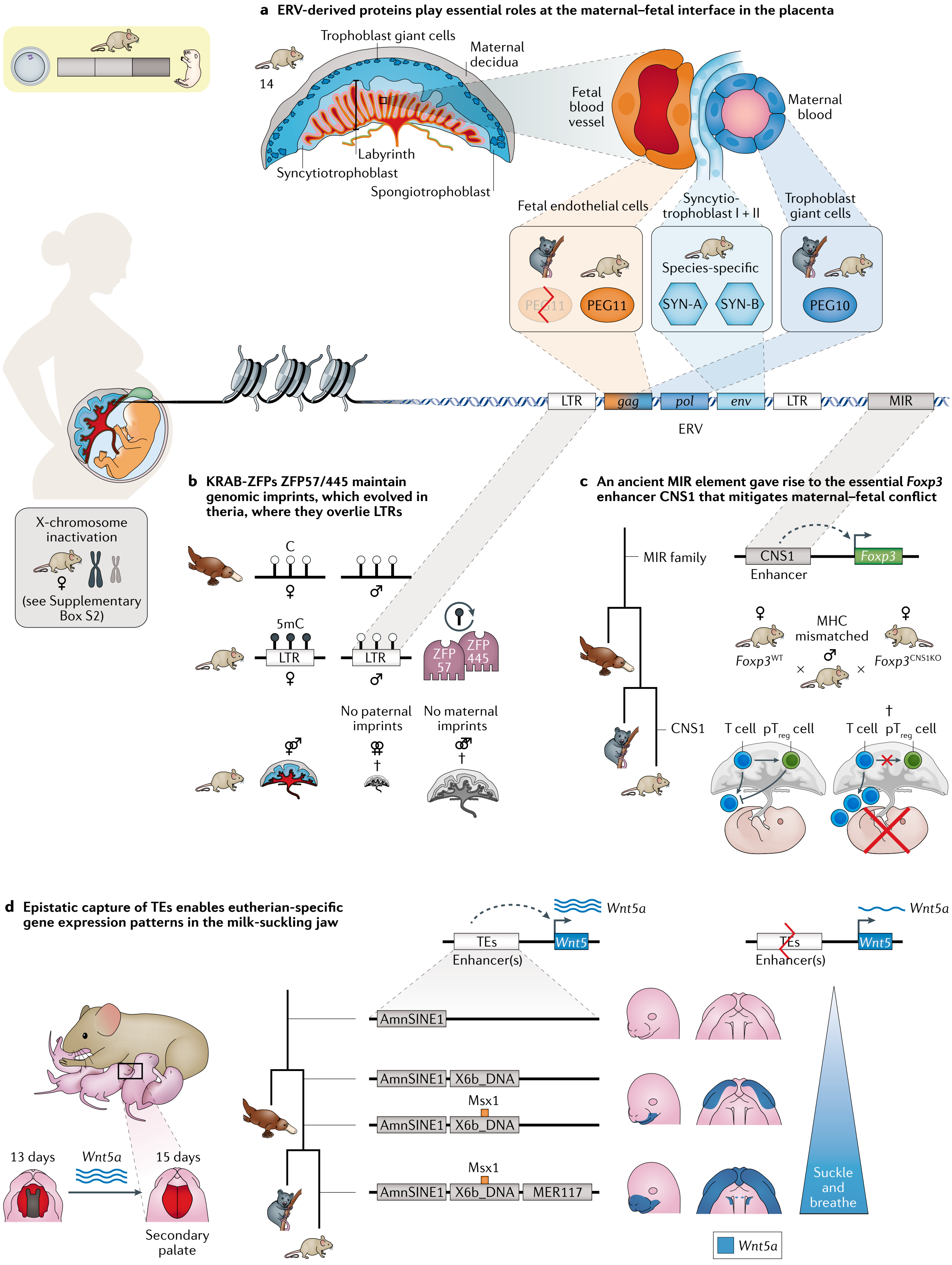
maternal–fetal interface, nutrient and gas exchange occurs via syncytiotrophoblast layers that form by cell fusion. In mice, this cell fusion depends on syncytin-A and syncytin-B that were independently co-opted from viral *env*-encoded proteins[152,153] (FIG. 4a). Many other therian species convergently acquired syncytin-like proteins from distinct ERV families that function to fuse trophoblast-derived cells, with critical gene sequences showing signs of purifying selection[154,155]. These include fusogenic human *SYNCYTIN-1* and *SYNCYTIN-2* that are implicated in placenta physiology and conserved in apes[156–160]. The function of env-derived syncytin genes is not limited to cell fusion. Syncytins also repress the antiviral host immune response, which was presumably necessary for them to enter the host[161]. The mammalian host, in turn, counteracts syncytin-mediated cell fusion. Amongst others, a truncated human endogenous retrovirus (HERV)-derived *env* with antiviral activity called SUPPRESSYN, also undergoing purifying selection in hominoids, inhibits SYNCYTIN 1-mediated cell fusion in human trophoblast cells[162–165]. *Env* proteins encoded by ERVs have thus repeatedly been co-opted for orchestrating trophoblast fusion across mammals. In mice, *Syncytin-A* has been shown to be indispensable for placental development and, consequently, the survival of embryos (see Supplementary Table S1).

Other ERV-encoded proteins, for example gag and pol, were similarly co-opted for essential placental functions. Two examples are the paternally expressed sushi-ichi retrotransposon-derived proteins PEG10 and PEG11 (REFS[166–169]) (see Supplementary Table S1). PEG10 is expressed both in the marsupial and eutherian placenta and in differentiating mouse TSCs[167,170]. In mice, *Peg10* is required for the formation of labyrinth and spongiotrophoblast layers of the placenta as well as for TSC differentiation in vitro[166,170–173]. PEG10 encapsulates, and thereby stabilizes, its own and other essential placental RNAs[170]. Mouse TSCs even bud off *Peg10*-containing vesicles, suggesting that PEG10 might mediate communication between cells via transfer of RNAs, similar to co-opted Ty3/gypsy *gag*-derived neuronal ARC in the vertebrate brain[170,174]. The related PEG11 prevents the syncytiotrophoblast from phagocytosing adjacent endothelial cells that line fetal capillaries in the chorioallantoic placenta in mice[175]. When *Peg11* is lost, clogged capillaries cause a variably poor fetal capillary network[176]. The gain of this eutherian-specific function (most marsupials do not have a chorioallantoic placenta) may have caused *Peg11* to be secondarily degraded in marsupials[177] (FIG. 4a). The different functions of PEG10 and PEG11 could be a direct consequence of their TE origin: whereas the *gag* domain of PEG10 harbours an RNA-binding site and the *pol* domain has a protease active site, the *gag*-encoded RNA-binding site is lacking from PEG11 (REF.[170]). Co-opted viral proteins thus represent key regulators of placental development that may have been crucial for live-born mammals to evolve.

Resolution of genetic conflict

With the advent of live birth, the therian fetus suddenly competed for resources with the pregnant mother, but not with its father. The resulting 'disagreement' of parental genomes over resources allocated for embryonic growth results in genomic conflict[178]. To resolve this conflict, the maternal and paternal genomes influence fetal growth via a process called genomic imprinting. Imprinting involves the selective epigenetic 'marking' of genes, in a parent of origin-specific manner, that regulate embryonic growth, often via the placenta, where consequently many imprinted genes are expressed[179]. How mammalian evolution was shaped by this genetic conflict becomes clear when imprinting is absent: whereas the placenta develops poorly in diploid gynogenetic mouse embryos, it overgrows in diploid androgenetic embryos, although neither survives to term[180,181] (FIG. 4b). This co-dependence of imprinting and successful development (particularly of the placenta) also explains why mammals always reproduce sexually, and never (except when critical imprinted genes such as H19 are mutated) parthenogenetically[182,183].

KRAB-ZFPs resolve genomic conflict by maintaining genomic imprinting. Following fertilization, mouse embryos undergo rapid genome-wide demethylation. 5mC is subsequently re-established in the embryonic lineage after implantation (and only weakly in the extra-embryonic tissues). Following gastrulation, the developing germ line undergoes genome-wide demethylation in a second wave of epigenetic reprogramming. Through the first wave of demethylation, a small number of imprinted loci (>100 in mice) retain parent-specific 5mC marks (imprints) leading to allele-specific expression. However, these imprints have to be erased and re-established according to sex in developing germ cells, so that sperm establish paternal imprints and developing oocytes establish maternal imprints. Strikingly, imprinting control regions (ICRs) often overlap with LTRs, for example mammalian apparent LTR retrotransposons (MaLRs) in mice (see Supplementary Box S1). Alternative promoters of non-canonically imprinted genes in the extra-embryonic tissues after implantation also overlie ERV-K LTRs whose imprints seem to depend on the H3K9me2 methyltransferase G9a (REFS[184,185]). These imprinted LTRs may have been important for the evolution of imprinting in mammals, as loci in the egg-laying platypus that became imprinted in theria harbour fewer TEs, and platypus overall has almost no LTRs[98,186] (FIGS 1a,4b). The KRAB-ZFPs ZFP445 and ZFP57 together maintain imprinting at all but one imprinted locus in mice (reviewed elsewhere[187])[188]. Consequently, their combined loss results in mid-gestation lethality[63,188] (see Supplementary Table S1). Although 80% of the binding sites of ZFP57 overlap with TEs, ZFP57 does not appear to play a role in TE silencing in mice[61]. Instead, ZFP57 maintains imprints by directly binding to a methylated hexanucleotide motif found at ICRs, to which it recruits KAP1–SETDB1 to deposit H3K9me3 and protects ICRs from passive and active TET-mediated DNA demethylation[58,189]. Whereas *Zfp57* is eutherian-specific, *Zfp445* has putative orthologues in marsupials, suggesting that ZFP445 is older and initially maintained imprinting. The only gene whose imprint is not maintained by ZFP57 or ZFP445 is the aforementioned



a ERV-derived proteins play essential roles at the maternal–fetal interface in the placenta
14
Trophoblast giant cells
Maternal decidua
Labyrinth
Syncytiotrophoblast
Spongiotrophoblast
Fetal blood vessel
Maternal blood
Fetal endothelial cells
Syncytio-trophoblast I + II
Trophoblast giant cells
PEG11
PEG11
Species-specific
SYN-A
SYN-B
PEG10
LTR
gag
pol
env
LTR
MIR
ERV
X-chromosome inactivation
♀
(see Supplementary Box S2)
b KRAB-ZFPs ZFP57/445 maintain genomic imprints, which evolved in theria, where they overlie LTRs
C
♀
♂
5mC
LTR
LTR
♀
♂
ZFP 57
ZFP 445
No paternal imprints
No maternal imprints
⚥
†
c An ancient MIR element gave rise to the essential Foxp3 enhancer CNS1 that mitigates maternal–fetal conflict
MIR family
CNS1
Enhancer
Foxp3
CNS1
MHC mismatched
Foxp3^WT
Foxp3^CNS1KO
×
T cell
pTreg cell
†
d Epistatic capture of TEs enables eutherian-specific gene expression patterns in the milk-suckling jaw
Wnt5a
TEs
Enhancer(s)
Wnt5
AmnSINE1
X6b_DNA
Msx1
MER117
Suckle and breathe
13 days
15 days
Secondary palate

Fig. 4 | **Co-option of TEs and regulatory mechanisms at the maternal–fetal interface.** **a** | In the mouse placenta, protein-coding endogenous retrovirus (ERV) sequences such as *gag*, *pol* and *env* (bottom) have been co-opted for essential functions at the maternal–fetal interface. At the maternal–fetal interface (top), trophoblast giant cell-lined maternal blood vessels (that allow the fetus to control blood supply) are in indirect contact with fetal endothelial cell-lined blood vessels. Indirect contact between maternal and fetal vessels is established via the syncytial syncytiotrophoblast (two layers in mice, one in human). The *env*-derived syncytins (in mouse SYN-A and SYN-B) mediate synytiotrophoblast cell fusion (which allows, for example, nutrient transfer) in mice. The *gag*/*pol*-derived PEG10 and *gag*-derived PEG11 are essential in trophoblast giant cells or fetal endothelial cells, respectively (see main text). **b** | Gynogenetic (♀♀) and androgenetic (♂♂) embryos lack imprinting and are not viable (†), among other reasons because the placenta undergrows (♀♀) or overgrows (♂♂). Maintenance of imprints at all but one imprinted gene in mice depend on the Krüppel-associated box zinc finger proteins (KRAB-ZFPs) ZFP57 and ZFP445. ZFP57 primarily binds (but does not repress) transposable element (TE) sequences. Loci that became imprinted in mice gained ERV sequences not present in monotremes (see FIG. 1a). **c** | Increased resorption of embryos in female mice that lack the *Foxp3* enhancer conserved non-coding sequence 1 (CNS1) that derives from a mammalian-wide interspersed repeat (MIR) element. Whereas the MIR family is found in all mammals, CNS1 colonized the *Foxp3* locus in eutheria. In the absence of CNS1, *Foxp3*-dependent peripheral regulatory T cell (pT_{reg} cell) differentiation does not occur. Instead, many more T cells than normal infiltrate the decidua, resulting in more fetuses being resorbed. Importantly, resorption of embryos only occurs when females are mated to immunologically distinct (major histocompatibility complex (MHC) mismatched) males, illustrating that CNS1 mitigates maternal–fetal conflict. **d** | Co-option of TEs from distinct classes gradually broadened the expression pattern of the *Wnt5* gene. WNT5 drives closure of the secondary palate, which enables simultaneous breathing and suckling, and thereby more efficient uptake of milk, but these TE-derived enhancers are not essential for this function. Given the contribution of large numbers of TEs (such as, for example, AmnSINE1 elements) to regulatory elements, epistatic captures of TEs such as this example may have more broadly influenced the evolution of novel gene expression patterns in mammals[204,205]. LTR, long terminal repeat; WT, wild type. Part **d** is adapted from REF.[203], CC BY 4.0 (https://creativecommons.org/licenses/by/4.0/).

Gynogenetic
Pertaining to gynogenesis, which is development whereby the embryo contains only maternal chromosomes. Biparental, that is diploid, gynogenetic embryos are produced by transplanting pronuclei between one-cell stage embryos to yield two female pronuclei.

Androgenetic
Pertaining to androgenesis, which is development whereby the embryo contains only paternal chromosomes (see gynogenetic).

Parthenogenetically
Pertaining to parthenogenesis, which is reproduction without fertilization ('virgin birth'). Parthenogenesis is common in invertebrates and plants.

Peg10, which may alternatively rely on a yet unidentified KRAB-ZFP[171,188]. Taken together, the evolution of genomic imprinting in theria may thus have relied on ancient TEs and TE-binding KRAB-ZFPs.

Linked with the evolution of genomic imprinting, theria evolved a second epigenetic process that is essential for placental development: X-chromosome inactivation[183,190,191]. Mary Lyon, who discovered X-chromosome inactivation in the 1960s, postulated that LINE elements, which are enriched on mammalian X chromosomes, are important for this process[192,193]. For a brief discussion of the relevance of TEs for the evolution of X-chromosome inactivation and mammalian development, we refer the interested reader to Supplementary Box S2.

Maternal adaptations to pregnancy

Mothers retain the embryo, nourish the fetus via the placenta and feed the newborn via breast milk, all while keeping themselves alive. The maternal endometrium, called the decidua during pregnancy, forms part of the eutherian placenta (BOX 3; FIGS 3,4). Decidualization is required for successful development and is, in part, responsible for age-related reproductive decline in mice[194]. Decidualization evolved in the common eutherian ancestor, probably from an inflammation and stress response to implantation and invasive placentation[195]. The immunological challenges that result from invasive placentation are known as maternal–fetal conflict and require immune tolerance by the mother.

TEs are essential to mitigate maternal–fetal conflict. To enforce maternal–fetal tolerance during pregnancy, eutherian-specific maternal peripheral regulatory T cells (pT_{reg} cells) infiltrate and accumulate in the mouse decidua. The differentiation of pT_{reg} cells from $CD4^+$ T cells requires the transcription factor FOXP3. *Foxp3* expression in pT_{reg} cells depends on a TGFβ-responsive enhancer called conserved non-coding sequence 1 (CNS1) that was co-opted from a specific tRNA-derived SINE retrotransposon family, the mammalian-wide interspersed repeats (MIRs)[196]. The MIR family probably colonized the common mammalian ancestor, but CNS1 inserted in the *Foxp3* locus in eutheria[196]. Studies of mice lacking CNS1 confirmed CNS1 as a *Foxp3* enhancer in pT_{reg} cells[196]. In the marsupial opossum and in zebrafish, ectopically inserted CNS1 sequences at *Foxp3* drove expression of a luciferase reporter, suggesting that its regulation is not eutheria-specific. In pregnant mice that lack CNS1, many more T cells than normal infiltrate the decidua, indicative of a systemic inflammatory response that resembles human pregnancy-associated disorders such as pre-eclampsia. As a result, maternal blood vessels are not properly remodelled and more fetuses are resorbed[196]. Strikingly, this increased resorption occurs only when females are mated to immunologically distinct allogeneic males, but not to immunologically identical syngeneic males[196] (FIG. 4c). In sum, this experiment demonstrated that MIR-derived CNS1 truly mitigates maternal–fetal conflict by enabling *Foxp3* expression in pT_{reg} cells specifically in eutheria.

The fetus, in turn, also mitigates maternal–fetal conflict via TE co-option. For example, several anthropoid primates bear a co-opted ERV1 element that acts as an enhancer to drive human leukocyte antigen G (HLA-G) expression, which induces immune tolerance in invasive trophoblast cells[197]. Similar to co-opted TEs in the mouse TSCs (see above), this ERV-derived trophoblast enhancer is positively regulated by critical trophectoderm regulators, in this case CEBPβ, GATA2 and GATA3 (REF.[197]). These examples suggest that TEs have played a vital role in modulating immune responses throughout the evolution of the eutherian maternal–fetal interface.

TEs underlie the decidual gene regulatory network. When decidualization evolved, MER20 elements colonized mammalian genomes[8,9]. MER20 elements are DNA transposons of the hobo Activator Tam3 (hAT) Charlie family that belongs to the medium reiteration frequency interspersed repeats (MERs)[8,9]. MER20 elements provided 13% of putative enhancers near genes that are both expressed in human decidual stromal cells and gained decidual expression in eutheria[8,9]. Another example of TE-driven gene expression in the decidua is the hormone prolactin that contributes to the proliferation and cytotoxicity of uterine natural killer cells that regulate placental invasion in humans and is essential for pregnancy in mice[198]. Distinct LTR-containing retrotransposons and L1 elements independently colonized the prolactin gene locus in different mammals: MER77 in mice, MER39 in primates and L1-2_LA

Mammalian apparent LTR retrotransposons
(MaLRs). Non-autonomous mouse-specific endogenous retroviruses (ERVs) that are the most common retroviral elements in the mouse, making up 4.8% of the genome, and are related to mouse and human ERV-L and human long terminal repeat (LTR) transposon-like human element 1 (THE1; an LTR-containing retrotransposon) elements.

AmnSINE1
A short interspersed nuclear element (SINE) family that amplified in the common amniote ancestor ~320 million years ago that has been conserved in ~200–700 copies in mouse and human genomes.

X-chromosome inactivation
The mammalian-specific process of almost complete silencing of one of the two X chromosomes to equalize the sex chromosome dose in female theria.

Decidualization
Changes of the maternal endometrium in preparation for, and during, mammalian pregnancy. It occurs during menstruation in humans, but is induced by pregnancy in mice (where it can also be triggered by uterine injection with mineral oil drops).

Peripheral regulatory T cells
(pT_{reg} cells). A FOXP3-expressing, specialized immunosuppressive T cell lineage generated extrathymically (peripheral). Regulatory T cells are also generated in the thymus (tT_{reg} cells). Regulatory T cells have essential functions in preventing fatal autoimmunity and inflammation and additional roles in metabolism and tissue repair.

Mammalian-wide interspersed repeats
(MIRs). A widespread subfamily of short interspersed nuclear elements (SINEs) that is often conserved in mammals.

Allogeneic
Genetically dissimilar, and hence immunologically incompatible (within a species); for example, major histocompatibility complex (MHC)-mismatched mouse strains. The opposite of syngeneic, which means genetically similar or identical.

in elephants[198]. MER77 and MER39 provided binding sites for the transcription factors SP1 and ETS1, respectively[198,199]. In apes, their synergistic action may have allowed particularly strong prolactin promoter activity[198,199]. These ERVs may thus have contributed to the more invasive placentation observed in apes, which may, amongst other factors, have enabled larger brain sizes. Similar to these diverse co-opted ancient TEs, the most ancient mammalian KRAB-ZFPs are also expressed in the human decidua and in immune cells[60]. Potentially reflecting the role of KRAB-ZFPs in decidualization, several KRAB-ZFP motifs are found within MER20 elements in human decidual stromal cells[60]. Both TEs and TE-binding KRAB-ZFPs are thus likely to have shaped the evolution of decidualization.

A THE1B element influences birth timing by regulating CRH expression. During human pregnancy, levels of syncytiotrophoblast-secreted corticotropin-releasing hormone (CRH) rise in the maternal blood and culminate just before birth[200]. Circulating CRH levels may serve as a 'placental clock' as they correlate with the onset of parturition. CRH placental expression in humans is driven at least in part by an LTR transposon-like human element 1B (THE1B) element[201]. Whereas most mammals express CRH exclusively in the brain, anthropoid primates use THE1B as a placenta-specific enhancer[201]. When a human *CRH* BAC transgene containing the THE1B element is introduced into mice, CRH becomes ectopically expressed in the placenta, and gestation lasts half a day longer. Reciprocally, subsequent THE1B deletion can revert ectopic THE1B-mediated prolonged gestation and placental (but not brain) CRH expression in mice[201]. Many other THE1B elements lie close to placentally expressed genes and influence their expression, and, more broadly, LTRs have generated placenta-expressed alternative transcripts in primates (for example, *CYP19*, *TMPRSS3* or *ENTPD1* (REF.[202])). Taken together, LTRs including THE1B elements may have influenced the evolution of primate pregnancy.

TEs enable eutherian-specific gene expression patterns in the milk-suckling jaw. Eutherian newborns can breathe and suckle simultaneously, whereas marsupials and monotremes cannot suckle milk as efficiently. To breathe and suckle at the same time, the oral and nasal cavity separate during eutherian development, which depends on the gene *Wnt5a*. Facial *Wnt5a* expression expanded successively during mammalian evolution by the co-option of three TEs that in reporter assays recapitulate ancestral *Wnt5* expression[203] (FIG. 4d). Upon deletion of these TEs, *Wnt5a* is abnormally expressed in mouse embryos, validating that these TEs contribute to the eutherian-specific gene expression patterns in the jaw[203]. Even though these TEs are required for neither the proper formation of the secondary palate nor survival of newborns, the *WNT5A* locus illustrates a common principle of how frequent epistatic capture of diverse TEs has shaped gene expression patterns throughout the evolution of mammalian development[203–205].

The known essential roles of TEs and KRAB-ZFPs at the maternal–fetal interface are among the best characterized and most convincing TE co-option events studied. TEs have been co-opted at all components of the maternal–fetal interface: in fetal blood vessels, at the syncytiotrophoblast and the maternal decidua, where they mediate nutrient and oxygen transfer and modulate immunological tolerance between fetus and mother. Particularly, the examples of essential ERV-derived proteins in the placenta demonstrate how the colonization of ERVs has directly impacted mammalian evolution. Similar to the earlier example of the TE-derived protein THAP11, at the maternal–fetal interface ERVs provide the raw genetic material that enables tissue interactions that are crucial both for the development and the evolution of the invasive placenta[115,116,149,151,206].

Conclusions and perspectives

The examples discussed in this Review illustrate how TEs can directly and indirectly regulate mammalian development. The vast majority of TE families, including many ERVs, seem to broadly, but subtly, influence species-specific gene expression in the form of transcripts and regulatory elements[207]. Particularly during early development, the concomitantly high expression of species-specific, often non-essential, KRAB-ZFP genes supports an ongoing arms race that may continue to promote genetic and regulatory innovation[26]. This arms race and the selfish nature of TEs may explain why so far tested TE-derived regulatory sequences prior to implantation are non-essential. However, in future, it is likely that more examples of essential TE functions will emerge. For example, a recent deletion of a mouse-specific MT2B2 element that acts as a mouse-specific alternative promoter of a conserved truncated CDK2AP1 isoform regulating cell proliferation affected pre-implantation development and reduced viability in 10-day-old pups (although the phenotype was only ~50% penetrant)[72]. Intriguingly, a distinct, more conserved TE drives truncated CDK2AP1 in other mammals, suggesting that mouse MT2B2 may act as a particularly potent driver of the isoform that may have accelerated pre-implantation development in mice[72].

The majority of known cases where TE co-option has resulted in an essential developmental process occur at the maternal–fetal interface[152,166,176,196,208]. As these traits are mammalian-specific, TEs may have critically influenced the evolution of mammalian development. Whether this conclusion holds true in mammals other than mice remains largely untested. Individual studies, such as the discovery of the aforementioned env-derived enJSRV sequence in sheep that is essential for trophoblast differentiation and successful development, support that other mammals, too, cannot develop without co-opted TE-derived sequences[208].

Even in mice, most TEs and KRAB-ZFPs are unstudied. KRAB-ZFPs are abundant, and a core set is conserved across all mammals with an invasive placenta. Two thirds of >80 screened embryonic lethal mouse knockouts from phenotyping consortia that arrest after mid-gestation have a dysmorphic placenta[209]. KAP1 has an essential but undescribed function in extra-embryonic tissues that is essential for gastrulation

Medium reiteration frequency interspersed repeats
(MERs). A loose grouping of diverse types of DNA transposons and retrotransposons. For example, whereas MER20 is a non-autonomous eutherian-specific subfamily of hobo Activator Tam3 (hAT) Charlie DNA transposons (~4,800 mouse and ~16,000 human copies), MER77 (~400 copies in mouse and 1,200 copies in human) and MER39 (~2600 human copies) are endogenous retroviruses (ERVs).

Parturition
The act or process of giving birth.

LTR transposon-like human element 1B
(THE1B). A long terminal repeat (LTR)-containing retrotransposon that colonized anthropoid primates ~50 million years ago and is present in ~20,000 copies in the human genome.

Epistatic
Pertaining to epistasis, which is the genetic phenomenon whereby the effect of a mutation depends on the presence or absence of other mutations.

MT2B2
A mouse-specific long terminal repeat (LTR) of a class 3 endogenous retrovirus (ERV) present in ~15,000 copies.

and is likely to involve the tissue that develops into the placenta. This finding implicates an unknown role for KRAB-ZFPs in extra-embryonic development. Intriguingly, loss of the epigenetic process of genomic imprinting, which depends on ZFP57 and ZFP445, leads to placental defects. The mammalian placenta and embryonic cell types form and may have evolved through interactions between embryonic, extra-embryonic and maternal tissues[115,116,149,151,206]. Even though TEs contribute to the function of all these tissues, it is unclear how TEs became integrated into the reciprocal signalling between them that drives development.

Metastatic cancers abundantly express TEs and TE-derived oncogenes and are hypomethylated in a similar way to invasive placentae[99,178,210]. Indeed, layers of fibroblasts from cows (where mothers suppress trophoblast invasion) are less susceptible to cancer cell invasion than layers of human fibroblasts[211]. Although TEs were not assessed in this experiment, downregulating genes that respond to trophoblast cells in human but not bovine fibroblasts confer human fibroblast layers a higher resistance to cancer cell invasion, suggesting that invasive placentation and cancer are causally linked[211]. Beyond cancer, neurodegeneration and autoimmune disease have also been linked to TEs or TE regulators[34,212]. It is tempting to speculate that females are particularly prone to these diseases because of TE-dependent female-specific adaptations to pregnancy, such as allowing invasive placentation and fetal immune tolerance[178,179,213,214].

Finally, invasive placentation is specific to live-born mammals, but placentation and genetic or maternal–fetal conflict are not[206]. Pregnant male sea horses activate gene expression profiles partially resembling immune responses in human pregnancy, illustrating that aspects of placentation convergently evolve[215]. In rare cases, the resolution of genetic conflicts or placentation outside mammals have been linked to TE co-option. Placental lizards, for example, encode a syncytin, and TEs in the extra-embryonic endosperm in *Arabidopsis thaliana* seeds drive genomic imprinting[216–218]. Undoubtedly, these examples of convergent evolution of TE co-option will teach us common principles about how TEs shape the evolution of development.

Addressing TE functions in mammalian development is challenging: dissecting early post-implantation-stage embryos requires extensive experience and cell numbers are insufficient for many biochemical assays. Although single-cell RNA-sequencing data sets of mammalian development become increasingly available (see, for example, Related links at the end of this Review), cell types in the developing placenta and maternal tissues are not yet well defined. Thus, current research mostly focuses on more easily accessible stem cells or pre-implantation embryos and embryonic rather than extra-embryonic development. As only a few knockouts have been studied, the essential roles of TEs and TE repression have remained controversial. Moreover, human embryonic, trophoblast or endometrial organoids, ESC-derived gastruloids or primordial germ cell-like cells and TSCs made human development accessible in vitro[219–229]. However, these in vitro systems are currently unable to undergo ordered gastrulation, organogenesis and placental development or simulate maternal responses to pregnancy as they occur in vivo. They cannot, therefore, replace phenotypic analysis of validated knockout mouse models.

Many effects of TE expression, repression and co-option are not well understood (see Supplementary Box S3). Given the rare yet substantial effects of key TEs on mammalian evolution and development, and the paucity of data on the vast majority of TEs and their regulators, including KRAB-ZFPs, this journey to discover our evolutionary origin will be an exciting one.

Published online: 05 August 2021

1. Lander, E. S. et al. Initial sequencing and analysis of the human genome. *Nature* **409**, 860–921 (2001).
2. Chimpanzee Sequencing and Analysis Consortium. Initial sequence of the chimpanzee genome and comparison with the human genome. *Nature* **437**, 69–87 (2005).
3. Waterston, R. H. et al. Initial sequencing and comparative analysis of the mouse genome. *Nature* **420**, 520–562 (2002).
4. Long, H. K., Prescott, S. L. & Wysocka, J. Ever-changing landscapes: transcriptional enhancers in development and evolution. *Cell* **167**, 1170–1187 (2016).
5. Wells, J. N. & Feschotte, C. A field guide to eukaryotic transposable elements. *Annu. Rev. Genet*. **54**, 539-561 (2020).
This article is a comprehensive introduction to features and distribution of major TE groups across eukaryote genomes in the context of their biology and evolution.
6. Smit, A. F. A., Hubley, R. & Green, P. RepeatMasker Open-4.0 (2013–2015).
7. Chuong, E. B., Rumi, M. A., Soares, M. J. & Baker, J. C. Endogenous retroviruses function as species-specific enhancer elements in the placenta. *Nat. Genet.* **45**, 325–329 (2013).
This study establishes that genome-wide species-specific gene regulatory networks in mouse TSCs rely on ERV-derived sequences, using the example of RLTR13D5 elements that provide binding sites for essential placental transcription factors.
8. Lynch, V. J. et al. Ancient transposable elements transformed the uterine regulatory landscape and transcriptome during the evolution of mammalian pregnancy. *Cell Rep.* **10**, 551–561 (2015).
9. Lynch, V. J., Leclerc, R. D., May, G. & Wagner, G. P. Transposon-mediated rewiring of gene regulatory networks contributed to the evolution of pregnancy in mammals. *Nat. Genet.* **43**, 1154–1159 (2011).
Using transcriptional profiling across mammalian endometrial cells, this work demonstrates that genes expressed in human endometrium gained endometrial expression through eutherian-specific MER20 elements that provide binding sites for transcription factors important for hormone responsiveness and pregnancy.
10. Trizzino, M. et al. Transposable elements are the primary source of novelty in primate gene regulation. *Genome Res.* **27**, 1623–1633 (2017).
11. Chuong, E. B., Elde, N. C. & Feschotte, C. Regulatory evolution of innate immunity through co-option of endogenous retroviruses. *Science* **351**, 1083–1087 (2016).
12. Bourque, G. et al. Ten things you should know about transposable elements. *Genome Biol.* **19**, 199 (2018).
This article is a useful primer by leaders in the field on the importance of mammalian TEs for genome evolution and function that introduces their properties and interactions with host cells.
13. Kazazian, H. H. Jr. & Moran, J. V. Mobile DNA in health and disease. *N. Engl. J. Med.* **377**, 361–370 (2017).
14. Deniz, O., Frost, J. M. & Branco, M. R. Regulation of transposable elements by DNA modifications. *Nat. Rev. Genet.* **20**, 417–431 (2019).
15. Cosby, R. L., Chang, N. C. & Feschotte, C. Host–transposon interactions: conflict, cooperation, and cooption. *Genes Dev.* **33**, 1098–1116 (2019).
This study uncovers ~100 fusion proteins combining DNA transposase and host-derived domains in tetrapod genomes, suggesting that novel mammalian transcriptional regulators can arise via this process, which is confirmed for the bat fusion protein KRABINER.
16. Bruno, M., Mahgoub, M. & Macfarlan, T. S. The arms race between KRAB-zinc finger proteins and endogenous retroelements and its impact on mammals. *Annu. Rev. Genet.* **53**, 393–416 (2019).
17. Imbeault, M., Helleboid, P. Y. & Trono, D. KRAB zinc-finger proteins contribute to the evolution of gene regulatory networks. *Nature* **543**, 550–554 (2017).
By identifying KRAB-ZFP genes in extant vertebrate genomes, performing extensive phylogenetic analysis of these KRAB-ZFPs that allowed dating their evolutionary origin and determining the binding sites of the majority of human KRAB-ZFPs, this study reveals the extent to which the KRAB-ZFP gene family interacts with TEs across vertebrate evolution.
18. Thomas, J. H. & Schneider, S. Coevolution of retroelements and tandem zinc finger genes. *Genome Res.* **21**, 1800–1812 (2011).
This work presents compelling evidence that infiltration with LTR retroelements drives the divergence of tandem zinc finger proteins (which include many KRAB-ZFPs) in vertebrate genomes by showing that the number of tandem zinc finger genes and LTR retroviral elements correlate.

19. Ecco, G., Imbeault, M. & Trono, D. KRAB zinc finger proteins. *Development* **144**, 2719–2729 (2017).
20. Sheng, G. & Foley, A. C. Diversification and conservation of the extraembryonic tissues in mediating nutrient uptake during amniote development. *Ann. N. Y. Acad. Sci.* **1271**, 97–103 (2012).
21. Rodriguez-Terrones, D. & Torres-Padilla, M. E. Nimble and ready to mingle: transposon outbursts of early development. *Trends Genet.* **34**, 806–820 (2018).
22. Rowe, H. M. & Trono, D. Dynamic control of endogenous retroviruses during development. *Virology* **411**, 273–287 (2011).
23. Loebel, D. A., Tsoi, B., Wong, N., O'Rourke, M. P. & Tam, P. P. Restricted expression of ETn-related sequences during post-implantation mouse development. *Gene Expr. Patterns* **4**, 467–471 (2004).
24. Macfarlan, T. S. et al. Embryonic stem cell potency fluctuates with endogenous retrovirus activity. *Nature* **487**, 57–63 (2012).
25. Boroviak, T. et al. Single cell transcriptome analysis of human, marmoset and mouse embryos reveals common and divergent features of preimplantation development. *Development* **145**, dev167833 (2018).
26. Wolf, G. et al. KRAB-zinc finger protein gene expansion in response to active retrotransposons in the murine lineage. *eLife* **9**, e56337 (2020).
27. Grow, E. J. et al. Intrinsic retroviral reactivation in human preimplantation embryos and pluripotent cells. *Nature* **522**, 221–225 (2015).
28. Liu, D. et al. Single-cell RNA-sequencing reveals the existence of naive and primed pluripotency in pre-implantation rhesus monkey embryos. *Genome Res.* **28**, 1481–1493 (2018).
29. Johnson, W. E. & Coffin, J. M. Constructing primate phylogenies from ancient retrovirus sequences. *Proc. Natl Acad. Sci. USA* **96**, 10254–10260 (1999).
30. Sundaram, V. et al. Widespread contribution of transposable elements to the innovation of gene regulatory networks. *Genome Res.* **24**, 1963–1976 (2014).
31. Goke, J. et al. Dynamic transcription of distinct classes of endogenous retroviral elements marks specific populations of early human embryonic cells. *Cell Stem Cell* **16**, 135–141 (2015).
32. Colombo, A. R., Triche, T. Jr. & Ramsingh, G. Transposable element expression in acute myeloid leukemia transcriptome and prognosis. *Sci. Rep.* **8**, 16449 (2018).
33. Kong, Y. et al. Transposable element expression in tumors is associated with immune infiltration and increased antigenicity. *Nat. Commun.* **10**, 5228 (2019).
34. Tam, O. H., Ostrow, L. W. & Gale Hammell, M. Diseases of the nERVous system: retrotransposon activity in neurodegenerative disease. *Mob. DNA* **10**, 32 (2019).
35. Babaian, A. & Mager, D. L. Endogenous retroviral promoter exaptation in human cancer. *Mob. DNA* **7**, 24 (2016).
36. Pasquesi, G. I. M. et al. Vertebrate lineages exhibit diverse patterns of transposable element regulation and expression across tissues. *Genome Biol. Evol.* **12**, 506–521 (2020).
37. Goerner-Potvin, P. & Bourque, G. Computational tools to unmask transposable elements. *Nat. Rev. Genet.* **19**, 688–704 (2018).
38. Todd, C. D., Deniz, O., Taylor, D. & Branco, M. R. Functional evaluation of transposable elements as enhancers in mouse embryonic and trophoblast stem cells. *eLife* **8**, e44344 (2019).
This study uses individual genetic deletions of candidate ERV-derived regulatory elements in mouse embryonic and TSCs, and a broader CRISPR interference screen that results in minimal changes in the expression of nearby genes, to demonstrate that many TE-derived enhancers do not influence nearby gene expression.
39. Fuentes, D. R., Swigut, T. & Wysocka, J. Systematic perturbation of retroviral LTRs reveals widespread long-range effects on human gene regulation. *eLife* **7**, e35989 (2018).
This paper uses an elegant novel CRISPR activation/interference screen in human EC cells to reveal significant contributions of hundreds of ape-specific LH5HS elements to the regulation of nearby genes.
40. McDole, K. et al. In toto imaging and reconstruction of post-implantation mouse development at the single-cell level. *Cell Level. Cell* **175**, 859–876.e33 (2018).
41. Pijuan-Sala, B., Guibentif, C. & Gottgens, B. Single-cell transcriptional profiling: a window into embryonic cell-type specification. *Nat. Rev. Mol. Cell Biol.* **19**, 399–412 (2018).
42. Frank, J. A. & Feschotte, C. Co-option of endogenous viral sequences for host cell function. *Curr. Opin. Virol.* **25**, 81–89 (2017).
43. Hancks, D. C. & Kazazian, H. H. Jr. Roles for retrotransposon insertions in human disease. *Mob. DNA* **7**, 9 (2016).
44. Payer, L. M. & Burns, K. H. Transposable elements in human genetic disease. *Nat. Rev. Genet.* **20**, 760–772 (2019).
45. Gilbert, N., Lutz-Prigge, S. & Moran, J. V. Genomic deletions created upon LINE-1 retrotransposition. *Cell* **110**, 315–325 (2002).
46. Rowe, H. M. et al. KAP1 controls endogenous retroviruses in embryonic stem cells. *Nature* **463**, 237–240 (2010).
This work first identifies the critical role of the KRAB zinc finger co-repressor KAP1 in repressing IAP elements and many other TEs in mouse embryos and ESCs.
47. Helleboid, P. Y. et al. The interactome of KRAB zinc finger proteins reveals the evolutionary history of their functional diversification. *EMBO J.* **38**, e101220 (2019).
48. Stoll, G. A. et al. Structure of KAP1 tripartite motif identifies molecular interfaces required for retroelement silencing. *Proc. Natl Acad. Sci. USA* **116**, 15042–15051 (2019).
49. Dodge, J. E., Kang, Y. K., Beppu, H., Lei, H. & Li, E. Histone H3-K9 methyltransferase ESET is essential for early development. *Mol. Cell Biol.* **24**, 2478–2486 (2004).
50. Schultz, D. C., Ayyanathan, K., Negorev, D., Maul, G. G. & Rauscher, F. J. 3rd SETDB1: a novel KAP-1-associated histone H3, lysine 9-specific methyltransferase that contributes to HP1-mediated silencing of euchromatic genes by KRAB zinc-finger proteins. *Genes Dev.* **16**, 919–932 (2002).
51. Quenneville, S. et al. The KRAB-ZFP/KAP1 system contributes to the early embryonic establishment of site-specific DNA methylation patterns maintained during development. *Cell Rep.* **2**, 766–773 (2012).
52. Walsh, C. P., Chaillet, J. R. & Bestor, T. H. Transcription of IAP endogenous retroviruses is constrained by cytosine methylation. *Nat. Genet.* **20**, 116–117 (1998).
This paper first describes the role of DNA methylation via the maintenance DNA methyltransferase DNMT1 in the repression of TEs (IAP elements) in mouse embryos.
53. Smith, Z. D. & Meissner, A. DNA methylation: roles in mammalian development. *Nat. Rev. Genet.* **14**, 204–220 (2013).
54. Jönsson, M. E. et al. Activation of neuronal genes via LINE-1 elements upon global DNA demethylation in human neural progenitors. *Nat. Commun.* **10**, 3182 (2019).
55. Bestor, T. H., Edwards, J. R. & Boulard, M. Notes on the role of dynamic DNA methylation in mammalian development. *Proc. Natl Acad. Sci. USA* **112**, 6796–6799 (2015).
56. Deaton, A. M. & Bird, A. CpG islands and the regulation of transcription. *Genes Dev.* **25**, 1010–1022 (2011).
57. Rosic, S. et al. Evolutionary analysis indicates that DNA alkylation damage is a byproduct of cytosine DNA methyltransferase activity. *Nat. Genet.* **50**, 452–459 (2018).
58. Quenneville, S. et al. In embryonic stem cells, ZFP57/KAP1 recognize a methylated hexanucleotide to affect chromatin and DNA methylation of imprinting control regions. *Mol. Cell* **44**, 361–372 (2011).
59. Boulard, M., Rucli, S., Edwards, J. R. & Bestor, T. H. Methylation-directed glycosylation of chromatin factors represses retrotransposon promoters. *Proc. Natl Acad. Sci. USA* **117**, 14292–14298 (2020).
60. Liu, H., Chang, L. H., Sun, Y., Lu, X. & Stubbs, L. Deep vertebrate roots for mammalian zinc finger transcription factor subfamilies. *Genome Biol. Evol.* **6**, 510–525 (2014).
61. Shi, H. et al. ZFP57 regulation of transposable elements and gene expression within and beyond imprinted domains. *Epigenetics Chromatin* **12**, 49 (2019).
62. Garcia-Garcia, M. J., Shibata, M. & Anderson, K. V. Chato, a KRAB zinc-finger protein, regulates convergent extension in the mouse embryo. *Development* **135**, 3053–3062 (2008).
63. Li, X. et al. A maternal-zygotic effect gene, Zfp57, maintains both maternal and paternal imprints. *Dev. Cell* **15**, 547–557 (2008).
This study first identifies the KRAB-ZFP ZFP57 as essential for maintaining the mono-allelic expression of many imprinted genes in mice.
64. Casademunt, E. et al. The zinc finger protein NRIF interacts with the neurotrophin receptor p75(NTR) and participates in programmed cell death. *EMBO J.* **18**, 6050–6061 (1999).
65. Eszter Posfai, J. P. S. Defining totipotency using criteria of increasing stringency. *biorxiv* https://doi.org/10.1101/2020.03.02.972893 (2020).
66. Torres-Padilla, M. E. On transposons and totipotency. *Philos. Trans. R. Soc. Lond. B Biol. Sci.* **375**, 20190339 (2020).
This review highlights and critically assesses the diverse roles of species-specific TEs in totipotency (part of a great collection of research and review articles on TEs based on the Royal Society discussion meeting 'Crossroads between Transposons and Gene Rregulation' held in May 2019).
67. De Iaco, A. et al. DUX-family transcription factors regulate zygotic genome activation in placental mammals. *Nat. Genet.* **49**, 941–945 (2017).
68. Leidenroth, A. et al. Evolution of DUX gene macrosatellites in placental mammals. *Chromosoma* **121**, 489–497 (2012).
69. De Iaco, A., Verp, S., Offner, S., Grun, D. & Trono, D. DUX is a non-essential synchronizer of zygotic genome activation. *Development* **147**, dev177725 (2020).
70. Jachowicz, J. W. et al. LINE-1 activation after fertilization regulates global chromatin accessibility in the early mouse embryo. *Nat. Genet.* **49**, 1502–1510 (2017).
Using TALE-based transcriptional silencing to modulate L1 transcription, this study finds that modulating L1 in zygotes (but not at the late two-cell stage) affects genome-wide chromatin accessibility and reduces progression to the blastocyst stage, suggesting an important role of L1 in mammalian pre-implantation development.
71. Hendrickson, P. G. et al. Conserved roles of mouse DUX and human DUX4 in activating cleavage-stage genes and MERVL/HERVL retrotransposons. *Nat. Genet.* **49**, 925–934 (2017).
72. Andrew Modzelewski, W. S. A species-specific retrotransposon drives a conserved Cdk2ap1 isoform essential for preimplantation development. biorxiv https://doi.org/10.1101/2021.03.24.436683 (2021).
73. Percharde, M. et al. A LINE1–nucleolin partnership regulates early development and ESC identity. *Cell* **174**, 391–405.e19 (2018).
74. Stojic, L. et al. Specificity of RNAi, LNA and CRISPRi as loss-of-function methods in transcriptional analysis. *Nucleic Acids Res.* **46**, 5950–5966 (2018).
75. Bassett, A. R. et al. Considerations when investigating lncRNA function in vivo. *eLife* **3**, e03058 (2014).
76. Korotkevich, E. et al. The apical domain is required and sufficient for the first lineage segregation in the mouse embryo. *Dev. Cell* **40**, 235–247.e7 (2017).
77. Nishioka, N. et al. The Hippo signaling pathway components Lats and Yap pattern Tead4 activity to distinguish mouse trophectoderm from inner cell mass. *Dev. Cell* **16**, 398–410 (2009).
78. Posfai, E. et al. Position- and Hippo signaling-dependent plasticity during lineage segregation in the early mouse embryo. *eLife* **6**, e22906 (2017).
79. Nelson, A. C., Mould, A. W., Bikoff, E. K. & Robertson, E. J. Mapping the chromatin landscape and Blimp1 transcriptional targets that regulate trophoblast differentiation. *Sci. Rep.* **7**, 6793 (2017).
80. Sun, M. A. et al. Deciphering the evolution of the transcriptional and regulatory landscape in human placenta. *bioRxiv* https://doi.org/10.1101/2020.09.11.289686 (2020).
81. Furukawa, S., Kuroda, Y. & Sugiyama, A. A comparison of the histological structure of the placenta in experimental animals. *J. Toxicol. Pathol.* **27**, 11–18 (2014).
82. Haig, D. Retroviruses and the placenta. *Curr. Biol.* **22**, R609–R613 (2012).
This review suggests that retroviruses evolved placental expression to facilitate retroviral transmission from mother to offspring and from offspring to mother, and that retroviruses and host defence systems contributed to the remarkable diversity among mammalian placentae.
83. Chazaud, C., Yamanaka, Y., Pawson, T. & Rossant, J. Early lineage segregation between epiblast and

primitive endoderm in mouse blastocysts through the Grb2–MAPK pathway. *Dev. Cell* **10**, 615–624 (2006).
84. Nichols, J. et al. Formation of pluripotent stem cells in the mammalian embryo depends on the POU transcription factor Oct4. *Cell* **95**, 379–391 (1998).
85. Niwa, H., Miyazaki, J. & Smith, A. G. Quantitative expression of Oct-3/4 defines differentiation, dedifferentiation or self-renewal of ES cells. *Nat. Genet.* **24**, 372–376 (2000).
86. Kunarso, G. et al. Transposable elements have rewired the core regulatory network of human embryonic stem cells. *Nat. Genet.* **42**, 631–634 (2010).
87. Bourque, G. et al. Evolution of the mammalian transcription factor binding repertoire via transposable elements. *Genome Res.* **18**, 1752–1762 (2008).
88. Marnetto, D. et al. Evolutionary rewiring of human regulatory networks by waves of genome expansion. *Am. J. Hum. Genet.* **102**, 207–218 (2018).
89. Hermant, C. & Torres-Padilla, M. E. TFs for TEs: the transcription factor repertoire of mammalian transposable elements. *Genes Dev.* **35**, 22–39 (2021).
90. Ganesh, S. & Svoboda, P. Retrotransposon-associated long non-coding RNAs in mice and men. *Pflug. Arch.* **468**, 1049–1060 (2016).
91. Koyanagi-Aoi, M. et al. Differentiation-defective phenotypes revealed by large-scale analyses of human pluripotent stem cells. *Proc. Natl Acad. Sci. USA* **110**, 20569–20574 (2013).
92. Lu, X. et al. The retrovirus HERVH is a long noncoding RNA required for human embryonic stem cell identity. *Nat. Struct. Mol. Biol.* **21**, 423–425 (2014).
93. Wang, J. et al. Primate-specific endogenous retrovirus-driven transcription defines naive-like stem cells. *Nature* **516**, 405–409 (2014).
94. Takahashi, K. et al. The pluripotent stem cell-specific transcript ESRG is dispensable for human pluripotency. *PLoS Genet.* **17**, e1009587 (2021).
95. Black, S. G. et al. Viral particles of endogenous betaretroviruses are released in the sheep uterus and infect the conceptus trophectoderm in a transspecies embryo transfer model. *J. Virol.* **84**, 9078–9085 (2010).
96. Lewis, S. H., Reynolds-Kohler, C., Fox, H. E. & Nelson, J. A. HIV-1 in trophoblastic and villous Hofbauer cells, and haematological precursors in eight-week fetuses. *Lancet* **335**, 565–568 (1990).
97. Bianchi, D. W. The inadvertent discovery of human fetal cell microchimerism. *Clin. Chem.* **64**, 1400–1401 (2018).
98. Haig, D. Transposable elements: self-seekers of the germline, team-players of the soma. *Bioessays* **38**, 1158–1166 (2016).
99. Smith, Z. D. et al. Epigenetic restriction of extraembryonic lineages mirrors the somatic transition to cancer. *Nature* **549**, 543–547 (2017).
100. Corsinotti, A. et al. Global and stage specific patterns of Krüppel-associated-box zinc finger protein gene expression in murine early embryonic cells. *PLoS ONE* **8**, e56721 (2013).
101. Rowe, H. M. et al. TRIM28 repression of retrotransposon-based enhancers is necessary to preserve transcriptional dynamics in embryonic stem cells. *Genome Res.* **23**, 452–461 (2013).
102. Pontis, J. et al. Hominoid-specific transposable elements and KZFPs facilitate human embryonic genome activation and control transcription in naive human ESCs. *Cell Stem Cell* **24**, 724–735.e5 (2019).
Based on the stage-specific repression of SVA-derived neuronal enhancers by hominoid-specific KRAB-ZFPs, this study suggests that KRAB-ZFPs may allow the tissue-specific expression of TE-derived regulatory elements, which may facilitate their co-option by repressing immediate early, potentially toxic effects of TEs early in development.
103. Roussigne, M. et al. The THAP domain: a novel protein motif with similarity to the DNA-binding domain of P element transposase. *Trends Biochem. Sci.* **28**, 66–69 (2003).
104. Clouaire, T. et al. The THAP domain of THAP1 is a large C2CH module with zinc-dependent sequence-specific DNA-binding activity. *Proc. Natl Acad. Sci. USA* **102**, 6907–6912 (2005).
105. Parker, J. B., Yin, H., Vinckevicius, A. & Chakravarti, D. Host cell factor-1 recruitment to E2F-bound and cell-cycle-control genes is mediated by THAP11 and ZNF143. *Cell Rep.* **9**, 967–982 (2014).
106. Parker, J. B., Palchaudhuri, S., Yin, H., Wei, J. & Chakravarti, D. A transcriptional regulatory role of the THAP11–HCF-1 complex in colon cancer cell function. *Mol. Cell Biol.* **32**, 1654–1670 (2012).
107. Campagne, S., Saurel, O., Gervais, V. & Milon, A. Structural determinants of specific DNA-recognition by the THAP zinc finger. *Nucleic Acids Res.* **38**, 3466–3476 (2010).
108. Bessière, D. et al. Structure-function analysis of the THAP zinc finger of THAP1, a large C2CH DNA-binding module linked to Rb/E2F pathways. *J. Biol. Chem.* **283**, 4352–4363 (2008).
109. Dejosez, M. et al. Ronin is essential for embryogenesis and the pluripotency of mouse embryonic stem cells. *Cell* **133**, 1162–1174 (2008).
110. Durruthy-Durruthy, J. et al. Spatiotemporal reconstruction of the human blastocyst by single-cell gene-expression analysis informs induction of naive pluripotency. *Dev. Cell* **38**, 100–115 (2016).
111. Dejosez, M. et al. Ronin/Hcf-1 binds to a hyperconserved enhancer element and regulates genes involved in the growth of embryonic stem cells. *Genes Dev.* **24**, 1479–1484 (2010).
112. Michaud, J. et al. HCFC1 is a common component of active human CpG-island promoters and coincides with ZNF143, THAP11, YY1, and GABP transcription factor occupancy. *Genome Res.* **23**, 907–916 (2013).
113. Mazars, R. et al. The THAP-zinc finger protein THAP1 associates with coactivator HCF-1 and O-GlcNAc transferase: a link between DYT6 and DYT3 dystonias. *J. Biol. Chem.* **285**, 13364–13371 (2010).
114. Cosby, R. L. et al. Recurrent evolution of vertebrate transcription factors by transposase capture. *Science* **371**, eabc6405 (2021).
115. Arnold, S. J. & Robertson, E. J. Making a commitment: cell lineage allocation and axis patterning in the early mouse embryo. *Nat. Rev. Mol. Cell Biol.* **10**, 91–103 (2009).
116. Bardot, E. S. & Hadjantonakis, A. K. Mouse gastrulation: coordination of tissue patterning, specification and diversification of cell fate. *Mech. Dev.* **163**, 103617 (2020).
117. Svoboda, P. et al. RNAi and expression of retrotransposons MuERV-L and IAP in preimplantation mouse embryos. *Dev. Biol.* **269**, 276–285 (2004).
118. Piko, L., Hammons, M. D. & Taylor, K. D. Amounts, synthesis, and some properties of intracisternal A particle-related RNA in early mouse embryos. *Proc. Natl Acad. Sci. USA* **81**, 488–492 (1984).
119. Poznanski, A. A. & Calarco, P. G. The expression of intracisternal A particle genes in the preimplantation mouse embryo. *Dev. Biol.* **143**, 271–281 (1991).
120. Auclair, G., Guibert, S., Bender, A. & Weber, M. Ontogeny of CpG island methylation and specificity of DNMT3 methyltransferases during embryonic development in the mouse. *Genome Biol.* **15**, 545 (2014).
121. Chapman, V., Forrester, L., Sanford, J., Hastie, N. & Rossant, J. Cell lineage-specific undermethylation of mouse repetitive DNA. *Nature* **307**, 284–286 (1984).
122. Brulet, P., Kaghad, M., Xu, Y. S., Croissant, O. & Jacob, F. Early differential tissue expression of transposon-like repetitive DNA sequences of the mouse. *Proc. Natl Acad. Sci. USA* **80**, 5641–5645 (1983).
123. Maksakova, I. A. & Mager, D. L. Transcriptional regulation of early transposon elements, an active family of mouse long terminal repeat retrotransposons. *J. Virol.* **79**, 13865–13874 (2005).
124. O'Donnell, K. A., An, W., Schrum, C. T., Wheelan, S. J. & Boeke, J. D. Controlled insertional mutagenesis using a LINE-1 (ORFeus) gene-trap mouse model. *Proc. Natl Acad. Sci. USA* **110**, E2706–E2713 (2013).
125. Wallace, N. A., Belancio, V. P. & Deininger, P. L. L1 mobile element expression causes multiple types of toxicity. *Gene* **419**, 75–81 (2008).
126. Cammas, F. et al. Mice lacking the transcriptional corepressor TIF1β are defective in early postimplantation development. *Development* **127**, 2955–2963 (2000).
127. Matsui, T. et al. Proviral silencing in embryonic stem cells requires the histone methyltransferase ESET. *Nature* **464**, 927–931 (2010).
128. Dai, H. Q. et al. TET-mediated DNA demethylation controls gastrulation by regulating Lefty–Nodal signalling. *Nature* **538**, 528–532 (2016).
129. Harten, S. K. et al. The first mouse mutants of D14Abb1e (Fam208a) show that it is critical for early development. *Mamm. Genome* **25**, 293–303 (2014).
130. Soehn, A. S. et al. Periphilin is strongly expressed in the murine nervous system and is indispensable for murine development. *Genesis* **47**, 697–707 (2009).
131. Zhu, Y., Wang, G. Z., Cingöz, O. & Goff, S. P. NP220 mediates silencing of unintegrated retroviral DNA. *Nature* **564**, 278–282 (2018).
132. Robbez-Masson, L. et al. The HUSH complex cooperates with TRIM28 to repress young retrotransposons and new genes. *Genome Res.* **28**, 836–845 (2018).
133. Rossant, J., Sanford, J. P., Chapman, V. M. & Andrews, G. K. Undermethylation of structural gene sequences in extraembryonic lineages of the mouse. *Dev. Biol.* **117**, 567–573 (1986).
134. Yuan, P. et al. Eset partners with Oct4 to restrict extraembryonic trophoblast lineage potential in embryonic stem cells. *Genes Dev.* **23**, 2507–2520 (2009).
135. Hemberger, M. Epigenetic landscape required for placental development. *Cell Mol. Life Sci.* **64**, 2422–2436 (2007).
136. Hemberger, M. Genetic–epigenetic intersection in trophoblast differentiation: implications for extraembryonic tissue function. *Epigenetics* **5**, 24–29 (2010).
137. Yang, X. et al. Nuclear reprogramming of cloned embryos and its implications for therapeutic cloning. *Nat. Genet.* **39**, 295–302 (2007).
138. Tanaka, S. et al. Placentomegaly in cloned mouse concepti caused by expansion of the spongiotrophoblast layer. *Biol. Reprod.* **65**, 1813–1821 (2001).
139. Ng, R. K. et al. Epigenetic restriction of embryonic cell lineage fate by methylation of Elf5. *Nat. Cell Biol.* **10**, 1280–1290 (2008).
140. Liu, S. et al. Setdb1 is required for germline development and silencing of H3K9me3-marked endogenous retroviruses in primordial germ cells. *Genes Dev.* **28**, 2041–2055 (2014).
141. Shibata, M., Blauvelt, K. E., Liem, K. F. Jr. & Garcia-Garcia, M. J. TRIM28 is required by the mouse KRAB domain protein ZFP568 to control convergent extension and morphogenesis of extra-embryonic tissues. *Development* **138**, 5333–5343 (2011).
This study, carefully characterizing conditional and knockout mouse embryos for *Trim28*, reveals that the essential role of ZFP568 during mouse gastrulation is partially phenocopied in conditional *Trim28* mutants, also establishing that KAP1 (encoded by *Trim28*) has crucial extra-embryonic roles.
142. Shibata, M. & Garcia-Garcia, M. J. The mouse KRAB zinc-finger protein CHATO is required in embryonic-derived tissues to control yolk sac and placenta morphogenesis. *Dev. Biol.* **349**, 331–341 (2011).
143. Yang, P. et al. A placental growth factor is silenced in mouse embryos by the zinc finger protein ZFP568. *Science* **356**, 757–759 (2017).
144. DeChiara, T. M., Efstratiadis, A. & Robertson, E. J. A growth-deficiency phenotype in heterozygous mice carrying an insulin-like growth factor II gene disrupted by targeting. *Nature* **345**, 78–80 (1990).
145. Sun, F. L., Dean, W. L., Kelsey, G., Allen, N. D. & Reik, W. Transactivation of Igf2 in a mouse model of Beckwith–Wiedemann syndrome. *Nature* **389**, 809–815 (1997).
146. Patel, A. et al. DNA conformation induces adaptable binding by tandem zinc finger proteins. *Cell* **173**, 221–233.e12 (2018).
147. Chien, H. C. et al. Targeted disruption in mice of a neural stem cell-maintaining, KRAB-Zn finger-encoding gene that has rapidly evolved in the human lineage. *PLoS ONE* **7**, e47481 (2012).
148. Ben-Haim, N. et al. The nodal precursor acting via activin receptors induces mesoderm by maintaining a source of its convertases and BMP4. *Dev. Cell* **11**, 313–323 (2006).
149. Robertson, E. J. Dose-dependent Nodal/Smad signals pattern the early mouse embryo. *Semin. Cell Dev. Biol.* **32**, 73–79 (2014).
150. Beddington, R. S. & Robertson, E. J. Axis development and early asymmetry in mammals. *Cell* **96**, 195–209 (1999).
151. Hemberger, M., Hanna, C. W. & Dean, W. Mechanisms of early placental development in mouse and humans. *Nat. Rev. Genet.* **21**, 27–43 (2020).
152. Dupressoir, A. et al. Syncytin-A knockout mice demonstrate the critical role in placentation of a fusogenic, endogenous retrovirus-derived, envelope gene. *Proc. Natl Acad. Sci. USA* **106**, 12127–12132 (2009).
Following the initial identification of syncytins in humans and mice in 2000 and 2005, this study first shows that loss of the endogenous retroviral env gene *Syncytin-A* in mice disables syncytiotrophoblast fusion, leading to embryonic lethality due to inability to establish the maternal–fetal interface.

153. Dupressoir, A. et al. A pair of co-opted retroviral envelope syncytin genes is required for formation of the two-layered murine placental syncytiotrophoblast. *Proc. Natl Acad. Sci. USA* **108**, E1164–E1173 (2011).
154. Lavialle, C. et al. Paleovirology of 'syncytins', retroviral env genes exapted for a role in placentation. *Philos. Trans. R. Soc. Lond. B Biol. Sci.* **368**, 20120507 (2013).
155. Cornelis, G. et al. Retroviral envelope syncytin capture in an ancestrally diverged mammalian clade for placentation in the primitive Afrotherian tenrecs. *Proc. Natl Acad. Sci. USA* **111**, E4332–E4341 (2014).
156. Herve, C. A., Forrest, G., Lower, R., Griffiths, D. J. & Venables, P. J. Conservation and loss of the ERV3 open reading frame in primates. *Genomics* **83**, 940–943 (2004).
157. de Parseval, N. et al. Comprehensive search for intra- and inter-specific sequence polymorphisms among coding envelope genes of retroviral origin found in the human genome: genes and pseudogenes. *BMC Genomics* **6**, 117 (2005).
158. Mallet, F. et al. The endogenous retroviral locus ERVWE1 is a bona fide gene involved in hominoid placental physiology. *Proc. Natl Acad. Sci. USA* **101**, 1731–1736 (2004).
159. de Parseval, N. & Heidmann, T. Physiological knockout of the envelope gene of the single-copy ERV-3 human endogenous retrovirus in a fraction of the Caucasian population. *J. Virol.* **72**, 3442–3445 (1998).
160. Kim, H. S. et al. Human endogenous retrovirus (HERV)-R family in primates: chromosomal location, gene expression, and evolution. *Gene* **370**, 34–42 (2006).
161. Mangeney, M. et al. Placental syncytins: genetic disjunction between the fusogenic and immunosuppressive activity of retroviral envelope proteins. *Proc. Natl Acad. Sci. USA* **104**, 20534–20539 (2007).
162. Sugimoto, J. et al. Suppressyn localization and dynamic expression patterns in primary human tissues support a physiologic role in human placentation. *Sci. Rep.* **9**, 19502 (2019).
163. Frank, J. A. Antiviral activity of a human placental protein of retroviral origin. *biorxiv* https://doi.org/10.1101/2020.08.23.263665 (2020).
164. Sugimoto, J., Sugimoto, M., Bernstein, H., Jinno, Y. & Schust, D. A novel human endogenous retroviral protein inhibits cell–cell fusion. *Sci. Rep.* **3**, 1462 (2013).
165. Buchrieser, J. et al. IFITM proteins inhibit placental syncytiotrophoblast formation and promote fetal demise. *Science* **365**, 176–180 (2019).
166. Ono, R. et al. Deletion of Peg10, an imprinted gene acquired from a retrotransposon, causes early embryonic lethality. *Nat. Genet.* **38**, 101–106 (2006).
This study first demonstrates that the PEG10 gene was co-opted from gag and pol proteins derived from ERVs to perform an essential placental function in mice.
167. Kaneko-Ishino, T. & Ishino, F. Mammalian-specific genomic functions: newly acquired traits generated by genomic imprinting and LTR retrotransposon-derived genes in mammals. *Proc. Jpn. Acad. Ser. B Phys. Biol. Sci.* **91**, 511–538 (2015).
168. Youngson, N. A., Kocialkowski, S., Peel, N. & Ferguson-Smith, A. C. A small family of sushi-class retrotransposon-derived genes in mammals and their relation to genomic imprinting. *J. Mol. Evol.* **61**, 481–490 (2005).
169. Henke, C. et al. Selective expression of sense and antisense transcripts of the sushi-ichi-related retrotransposon-derived family during mouse placentogenesis. *Retrovirology* **12**, 9 (2015).
170. Abed, M. et al. The Gag protein PEG10 binds to RNA and regulates trophoblast stem cell lineage specification. *PLoS ONE* **14**, e0214110 (2019).
171. Koppes, E., Himes, K. P. & Chaillet, J. R. Partial loss of genomic imprinting reveals important roles for Kcnq1 and Peg10 imprinted domains in placental development. *PLoS ONE* **10**, e0135202 (2015).
172. Suzuki, S. et al. Retrotransposon silencing by DNA methylation can drive mammalian genomic imprinting. *PLoS Genet.* **3**, e55 (2007).
173. Ono, R. et al. A retrotransposon-derived gene, PEG10, is a novel imprinted gene located on human chromosome 7q21. *Genomics* **73**, 232–237 (2001).
174. Hantak, M. P., Einstein, J., Kearns, R. B. & Shepherd, J. D. Intercellular communication in the nervous system goes viral. *Trends Neurosci.* **44**, 248–259 (2021).
175. Kitazawa, M., Tamura, M., Kaneko-Ishino, T. & Ishino, F. Severe damage to the placental fetal capillary network causes mid- to late fetal lethality and reduction in placental size in Peg11/Rtl1 KO mice. *Genes. Cell* **22**, 174–188 (2017).
176. Sekita, Y. et al. Role of retrotransposon-derived imprinted gene, Rtl1, in the feto-maternal interface of mouse placenta. *Nat. Genet.* **40**, 243–248 (2008).
177. Edwards, C. A. et al. The evolution of the DLK1–DIO3 imprinted domain in mammals. *PLoS Biol.* **6**, e135 (2008).
178. Haig, D. Maternal–fetal conflict, genomic imprinting and mammalian vulnerabilities to cancer. *Philos. Trans. R. Soc. Lond. B Biol. Sci.* **370**, 20140178 (2015).
179. Haig, D. Going retro: transposable elements, embryonic stem cells, and the mammalian placenta (retrospective on DOI 10.1002/bies.201300059). *Bioessays* **37**, 1154 (2015).
180. McGrath, J. & Solter, D. Completion of mouse embryogenesis requires both the maternal and paternal genomes. *Cell* **37**, 179–183 (1984).
181. Surani, M. A., Barton, S. C. & Norris, M. L. Nuclear transplantation in the mouse: heritable differences between parental genomes after activation of the embryonic genome. *Cell* **45**, 127–136 (1986).
182. Kono, T. et al. Birth of parthenogenetic mice that can develop to adulthood. *Nature* **428**, 860–864 (2004).
183. Reik, W. & Lewis, A. Co-evolution of X-chromosome inactivation and imprinting in mammals. *Nat. Rev. Genet.* **6**, 403–410 (2005).
184. Hanna, C. W. et al. Endogenous retroviral insertions drive non-canonical imprinting in extra-embryonic tissues. *Genome Biol.* **20**, 225 (2019).
185. Andergassen, D., Smith, Z. D., Rinn, J. L. & Meissner, A. Diverse mechanisms for epigenetic imprinting in mammals. *bioRxiv* https://doi.org/10.1101/2021.04.30.442087f (2021).
186. Pask, A. J. et al. Analysis of the platypus genome suggests a transposon origin for mammalian imprinting. *Genome Biol.* **10**, R1 (2009).
187. Juan, A. M. & Bartolomei, M. S. Evolving imprinting control regions: KRAB zinc fingers hold the key. *Genes Dev.* **33**, 1–3 (2019).
188. Takahashi, N. et al. ZNF445 is a primary regulator of genomic imprinting. *Genes Dev.* **33**, 49–54 (2019).
This study establishes that the essential KRAB-ZFP ZFP57 pairs with ZNF445 to maintain all but one (Peg10) ICRs in mice.
189. Alexander, K. A., Wang, X., Shibata, M., Clark, A. G. & García-García, M. J. TRIM28 controls genomic imprinting through distinct mechanisms during and after early genome-wide reprogramming. *Cell Rep.* **13**, 1194–1205 (2015).
190. Mugford, J. W., Yee, D. & Magnuson, T. Failure of extra-embryonic progenitor maintenance in the absence of dosage compensation. *Development* **139**, 2130–2138 (2012).
191. Marahrens, Y., Panning, B., Dausman, J., Strauss, W. & Jaenisch, R. Xist-deficient mice are defective in dosage compensation but not spermatogenesis. *Genes Dev.* **11**, 156–166 (1997).
192. Lyon, M. F. Gene action in the X-chromosome of the mouse (*Mus musculus* L.). *Nature* **190**, 372–373 (1961).
In this paper, Mary Lyon — who discovered the epigenetic process of X-chromosome inactivation — suggests that LINE elements, which are enriched on mammalian X chromosomes, are important for this process.
193. Lyon, M. F. X-chromosome inactivation: a repeat hypothesis. *Cytogenet. Cell Genet.* **80**, 133–137 (1998).
194. Woods, L. et al. Decidualisation and placentation defects are a major cause of age-related reproductive decline. *Nat. Commun.* **8**, 352 (2017).
195. Erkenbrack, E. M. et al. The mammalian decidual cell evolved from a cellular stress response. *PLoS Biol.* **16**, e2005594 (2018).
196. Samstein, R. M., Josefowicz, S. Z., Arvey, A., Treuting, P. M. & Rudensky, A. Y. Extrathymic generation of regulatory T cells in placental mammals mitigates maternal-fetal conflict. *Cell* **150**, 29–38 (2012).
This article demonstrates using sophisticated allogenic and syngeneic mouse models that the lack of the eutherian-specific MIR element-derived Foxp3 enhancer plays an essential role in mitigating maternal–fetal conflict, thereby providing a compelling example that a single TE-derived regulatory sequence can be essential for mammalian development.
197. Ferreira, L. M. et al. A distant trophoblast-specific enhancer controls HLA-G expression at the maternal-fetal interface. *Proc. Natl Acad. Sci. USA* **113**, 5364–5369 (2016).
198. Emera, D. et al. Convergent evolution of endometrial prolactin expression in primates, mice, and elephants through the independent recruitment of transposable elements. *Mol. Biol. Evol.* **29**, 239–247 (2012).
199. Emera, D. & Wagner, G. P. Transformation of a transposon into a derived prolactin promoter with function during human pregnancy. *Proc. Natl Acad. Sci. USA* **109**, 11246–11251 (2012).
200. McLean, M. et al. A placental clock controlling the length of human pregnancy. *Nat. Med.* **1**, 460–463 (1995).
201. Dunn-Fletcher, C. E. et al. Anthropoid primate-specific retroviral element THE1B controls expression of CRH in placenta and alters gestation length. *PLoS Biol.* **16**, e2006337 (2018).
202. van de Lagemaat, L. N., Landry, J. R., Mager, D. L. & Medstrand, P. Transposable elements in mammals promote regulatory variation and diversification of genes with specialized functions. *Trends Genet.* **19**, 530–536 (2003).
203. Nishihara, H. et al. Coordinately co-opted multiple transposable elements constitute an enhancer for wnt5a expression in the mammalian secondary palate. *PLoS Genet.* **12**, e1006380 (2016).
204. Nishihara, H., Smit, A. F. & Okada, N. Functional noncoding sequences derived from SINEs in the mammalian genome. *Genome Res.* **16**, 864–874 (2006).
205. Hirakawa, M., Nishihara, H., Kanehisa, M. & Okada, N. Characterization and evolutionary landscape of AmnSINE1 in amniota genomes. *Gene* **441**, 100–110 (2009).
206. Griffith, O. W. & Wagner, G. P. The placenta as a model for understanding the origin and evolution of vertebrate organs. *Nat. Ecol. Evol.* **1**, 72 (2017).
207. Sundaram, V. & Wysocka, J. Transposable elements as a potent source of diverse *cis*-regulatory sequences in mammalian genomes. *Philos. Trans. R. Soc. Lond. B Biol. Sci.* **375**, 20190347 (2020).
208. Dunlap, K. A. et al. Endogenous retroviruses regulate periimplantation placental growth and differentiation. *Proc. Natl Acad. Sci. USA* **103**, 14390–14395 (2006).
This study shows that knock down of the sheep env-derived enJSRVs leads to failure to differentiate binucleate trophoblast giant cells and lethality, providing compelling evidence that ERVs in mammals other than mice have co-opted TEs for essential developmental functions.
209. Perez-Garcia, V. et al. Placentation defects are highly prevalent in embryonic lethal mouse mutants. *Nature* **555**, 463–468 (2018).
210. D'Souza, A. W. & Wagner, G. P. Malignant cancer and invasive placentation: a case for positive pleiotropy between endometrial and malignancy phenotypes. *Evol. Med. Public Health* **2014**, 136–145 (2014).
211. Kshitiz et al. Evolution of placental invasion and cancer metastasis are causally linked. *Nat. Ecol. Evol.* **3**, 1743–1753 (2019).
212. Treger, R. S. et al. The lupus susceptibility locus Sgp3 encodes the suppressor of endogenous retrovirus expression SNERV. *Immunity* **50**, 334–347.e9 (2019).
213. Natri, H., Garcia, A. R., Buetow, K. H., Trumble, B. C. & Wilson, M. A. The pregnancy pickle: evolved immune compensation due to pregnancy underlies sex differences in human diseases. *Trends Genet.* **35**, 478–488 (2019).
214. Greenbaum, S. & Greenbaum, G. Evolution of immune sexual dimorphism in response to placental invasiveness: a response to Natri et al. *Trends Genet.* **36**, 3–5 (2019).
215. Roth, O. et al. Evolution of male pregnancy associated with remodeling of canonical vertebrate immunity in seahorses and pipefishes. *Proc. Natl Acad. Sci. USA* **117**, 9431–9439 (2020).
216. Cornelis, G. et al. An endogenous retroviral envelope syncytin and its cognate receptor identified in the viviparous placental Mabuya lizard. *Proc. Natl Acad. Sci. USA* **114**, E10991–E11000 (2017).
217. Gehring, M. & Satyaki, P. R. Endosperm and imprinting, inextricably linked. *Plant. Physiol.* **173**, 143–154 (2017).
218. Pignatta, D., Novitzky, K., Satyaki, P. R. V. & Gehring, M. A variably imprinted epiallele impacts seed development. *PLoS Genet.* **14**, e1007469 (2018).
219. Turco, M. Y. et al. Trophoblast organoids as a model for maternal–fetal interactions during human placentation. *Nature* **564**, 263–267 (2018).

220. Turco, M. Y. et al. Long-term, hormone-responsive organoid cultures of human endometrium in a chemically defined medium. *Nat. Cell Biol.* **19**, 568–577 (2017).
221. Mirna Marinic, V. J. L. Derivation of endometrial gland organoids from term post-partum placenta. *biorxiv* https://doi.org/10.1101/753780v1 (2019).
222. Beccari, L. et al. Multi-axial self-organization properties of mouse embryonic stem cells into gastruloids. *Nature* **562**, 272–276 (2018).
223. Yoney, A. et al. WNT signaling memory is required for ACTIVIN to function as a morphogen in human gastruloids. *eLife* **7**, e38279 (2018).
224. Hayashi, K. et al. Offspring from oocytes derived from in vitro primordial germ cell-like cells in mice. *Science* **338**, 971–975 (2012).
225. Hayashi, K., Ohta, H., Kurimoto, K., Aramaki, S. & Saitou, M. Reconstitution of the mouse germ cell specification pathway in culture by pluripotent stem cells. *Cell* **146**, 519–532 (2011).
226. Yamashiro, C. et al. Generation of human oogonia from induced pluripotent stem cells in vitro. *Science* **362**, 356–360 (2018).
227. Okae, H. et al. Derivation of human trophoblast stem cells. *Cell Stem Cell* **22**, 50–63.e6 (2018).
228. Haider, S. et al. Self-renewing trophoblast organoids recapitulate the developmental program of the early human placenta. *Stem Cell Rep.* **11**, 537–551 (2018).
229. Marinic, M., Rana, S. & Lynch, V. J. Derivation of endometrial gland organoids from term placenta. *Placenta* **101**, 75–79 (2020).
230. Wolf, D. & Goff, S. P. Embryonic stem cells use ZFP809 to silence retroviral DNAs. *Nature* **458**, 1201–1204 (2009).
231. Loebel, D. A., O'Rourke, M. P., Steiner, K. A., Banyer, J. & Tam, P. P. Isolation of differentially expressed genes from wild-type and Twist mutant mouse limb buds. *Genesis* **33**, 103–113 (2002).
232. Chen, Z. F. & Behringer, R. R. Twist is required in head mesenchyme for cranial neural tube morphogenesis. *Genes Dev.* **9**, 686–699 (1995).
233. Gray, P. A. et al. Mouse brain organization revealed through direct genome-scale TF expression analysis. *Science* **306**, 2255–2257 (2004).
234. Lu, X. et al. Twist1 is involved in trophoblast syncytialization by regulating GCM1. *Placenta* **39**, 45–54 (2016).
235. Kruger, I. et al. Sp1/Sp3 compound heterozygous mice are not viable: impaired erythropoiesis and severe placental defects. *Dev. Dyn.* **236**, 2235–2244 (2007).
236. Sawyer, M. H., Nachlas, N. E. Jr. & Panem, S. C-type viral antigen expression in human placenta. *Nature* **275**, 62–64 (1978).
237. Tsumura, A. et al. Maintenance of self-renewal ability of mouse embryonic stem cells in the absence of DNA methyltransferases Dnmt1, Dnmt3a and Dnmt3b. *Genes Cell* **11**, 805–814 (2006).
238. Argelaguet, R. et al. Multi-omics profiling of mouse gastrulation at single-cell resolution. *Nature* **576**, 487–491 (2019).
239. Mochizuki, K. et al. SETDB1 is essential for mouse primordial germ cell fate determination by ensuring BMP signaling. *Development* **145**, dev164160 (2018).
240. Zhu, D. et al. Lysine-specific demethylase 1 regulates differentiation onset and migration of trophoblast stem cells. *Nat. Commun.* **5**, 3174 (2014).
241. Herzog, M. et al. TIF1β association with HP1 is essential for post-gastrulation development, but not for Sertoli cell functions during spermatogenesis. *Dev. Biol.* **350**, 548–558 (2011).
242. McClintock, C. B. The origin and behavior of mutable loci in maize. *Proc. Natl Acad. Sci. USA* **36**, 344–355 (1950).
243. Comfort, N. C. From controlling elements to transposons: Barbara McClintock and the Nobel Prize. *Trends Biochem. Sci.* **26**, 454–457 (2001).
This paper is a brief historical perspective on Barbara McClintock's discovery of TEs in maize.
244. Feschotte, C. & Pritham, E. J. DNA transposons and the evolution of eukaryotic genomes. *Annu. Rev. Genet.* **41**, 331–368 (2007).
245. Zimmerly, S. & Semper, C. Evolution of group II introns. *Mob. DNA* **6**, 7 (2015).
246. Koonin, E. V. Viruses and mobile elements as drivers of evolutionary transitions. *Philos. Trans. R Soc. Lond. B Biol. Sci.* **371**,, 20150442 (2016).
247. Novikova, O. & Belfort, M. Mobile group II introns as ancestral eukaryotic elements. *Trends Genet.* **33**, 773–783 (2017).
248. Kramerov, D. A. & Vassetzky, N. S. Short retroposons in eukaryotic genomes. *Int. Rev. Cytol.* **247**, 165–221 (2005).
249. Malik, H. S., Burke, W. D. & Eickbush, T. H. The age and evolution of non-LTR retrotransposable elements. *Mol. Biol. Evol.* **16**, 793–805 (1999).
250. Krupovic, M., Dolja, V. V. & Koonin, E. V. Origin of viruses: primordial replicators recruiting capsids from hosts. *Nat. Rev. Microbiol.* **17**, 449–458 (2019).
251. Dewannieux, M., Esnault, C. & Heidmann, T. LINE-mediated retrotransposition of marked Alu sequences. *Nat. Genet.* **35**, 41–48 (2003).
252. Muñoz-Lopez, M., et al. LINE-1 retrotransposition impacts the genome of human pre-implantation embryos and extraembryonic tissues. *biorxiv* https://www.biorxiv.org/content/10.1101/522623v1 (2019).
253. Gagnier, L., Belancio, V. P. & Mager, D. L. Mouse germ line mutations due to retrotransposon insertions. *Mob. DNA* **10**, 15 (2019).
254. Richardson, S. R. et al. Heritable L1 retrotransposition in the mouse primordial germline and early embryo. *Genome Res.* **27**, 1395–1405 (2017).
255. Nellaker, C. et al. The genomic landscape shaped by selection on transposable elements across 18 mouse strains. *Genome Biol.* **13**, R45 (2012).
256. Gilbert, C. & Feschotte, C. Horizontal acquisition of transposable elements and viral sequences: patterns and consequences. *Curr. Opin. Genet. Dev.* **49**, 15–24 (2018).
257. Tarlinton, R. E., Meers, J. & Young, P. R. Retroviral invasion of the koala genome. *Nature* **442**, 79–81 (2006).
258. Birtle, Z. & Ponting, C. P. Meisetz and the birth of the KRAB motif. *Bioinformatics* **22**, 2841–2845 (2006).
This study traces the origin of the KRAB motif that characterizes KRAB zinc finger genes to sea urchins and presents evidence that a KRAB motif progenitor was present in the last common ancestor of animals, plants and fungi.
259. Najafabadi, H. S. et al. C2H2 zinc finger proteins greatly expand the human regulatory lexicon. *Nat. Biotechnol.* **33**, 555–562 (2015).
260. Jumpei Ito, I. K. et al. Endogenous retroviruses drive KRAB zinc-finger family protein expression for tumor suppression. *biorxiv* https://doi.org/10.1101/2020.02.02.931501 (2020).
261. Ivanov, D., Stone, J. R., Maki, J. L., Collins, T. & Wagner, G. Mammalian SCAN domain dimer is a domain-swapped homolog of the HIV capsid C-terminal domain. *Mol. Cell* **17**, 137–143 (2005).
262. Williams, A. J., Blacklow, S. C. & Collins, T. The zinc finger-associated SCAN box is a conserved oligomerization domain. *Mol. Cell Biol.* **19**, 8526–8535 (1999).
263. Al Chiblak, M., Steinbeck, F., Thiesen, H. J. & Lorenz, P. DUF3669, a "domain of unknown function" within ZNF746 and ZNF777, oligomerizes and contributes to transcriptional repression. *BMC Mol. Cell Biol.* **20**, 60 (2019).
264. Emerson, R. O. & Thomas, J. H. Gypsy and the birth of the SCAN domain. *J. Virol.* **85**, 12043–12052 (2011).
265. Jorgensen, P. Yolk. *Curr. Biol.* **18**, R103–R104 (2008).
266. Arendt, D. & Nubler-Jung, K. Rearranging gastrulation in the name of yolk: evolution of gastrulation in yolk-rich amniote eggs. *Mech. Dev.* **81**, 3–22 (1999).
267. Griffith, O. W. et al. Embryo implantation evolved from an ancestral inflammatory attachment reaction. *Proc. Natl Acad. Sci. USA* **114**, E6566–E6575 (2017).
268. Ross, C. & Boroviak, T. E. Origin and function of the yolk sac in primate embryogenesis. *Nat. Commun.* **11**, 3760 (2020).
269. Pereira, P. N. et al. Amnion formation in the mouse embryo: the single amniochorionic fold model. *BMC Dev. Biol.* **11**, 48 (2011).
270. Renfree, M. B. Review: Marsupials: placental mammals with a difference. *Placenta* **31** (Suppl.), S21–S26 (2010).
271. Rodriguez, T. A., Srinivas, S., Clements, M. P., Smith, J. C. & Beddington, R. S. Induction and migration of the anterior visceral endoderm is regulated by the extra-embryonic ectoderm. *Development* **132**, 2513–2520 (2005).
272. Waldrip, W. R., Bikoff, E. K., Hoodless, P. A., Wrana, J. L. & Robertson, E. J. Smad2 signaling in extraembryonic tissues determines anterior–posterior polarity of the early mouse embryo. *Cell* **92**, 797–808 (1998).
273. Lawson, K. A. et al. Bmp4 is required for the generation of primordial germ cells in the mouse embryo. *Genes Dev.* **13**, 424–436 (1999).
274. Martin, J. F. & Wagner, G. P. The origin of platelets enabled the evolution of eutherian placentation. *Biol. Lett.* **15**, 20190374 (2019).
275. Roberts, R. M., Green, J. A. & Schulz, L. C. The evolution of the placenta. *Reproduction* **152**, R179–189 (2016).

Acknowledgements
The Macfarlan laboratory is funded by the National Institutes of Health (NIH) (DIR 1ZIAHD008933) with additional funding through the Eunice Kennedy Shriver National Institute of Child Health and Human Development (NICHD) human placenta project. The authors thank L. Robertson, P. Kurbel, C. Simon, M. Almeida and J. Frank, and members of the Macfarlan laboratory as well as the NICHD Genetics and Epigenetics of Development Group, for discussions and critical comments on the manuscript. Special thanks to K. Pfeifer, P. Rocha, J. Thompson, A. Ivanoff, R. Cosby and M. Gauchier as well as the reviewers for their critical feedback. They apologize to those authors not cited due to space constraints.

Author contributions
Both authors contributed equally to all aspects of the article.

Competing interests
The authors declare no competing interests.

Peer review information
Nature Reviews Genetics thanks M. Branco, J. Rossant and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

Publisher's note
Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Supplementary information
The online version contains supplementary material available at https://doi.org/10.1038/s41576-021-00385-1.

RELATED LINKS
Dfam TE Classification: https://www.dfam.org/classification/
Dfam TE database: https://www.dfam.org/home
Endoderm Explorer (GUI mouse scRNA-seq): https://endoderm-explorer.com/
gnomAD human sequencing data (including pLi): https://gnomad.broadinstitute.org/
Gypsy TE Database (GyDB): http://gydb.org/index.php/Main_Page
Human Gastrulation (GUI human scRNA-seq): http://human-gastrula.net/
Maternal–Fetal interface (GUI human scRNA-seq): https://maternal-fetal-interface.cellgeni.sanger.ac.uk/
Mouse Genome Informatics (MGI): http://www.informatics.jax.org/
Mouse gastrulation (GUI mouse scRNA-seq): https://marionilab.cruk.cam.ac.uk/MouseGastrulation2018/
Pfam protein domain database: https://pfam.xfam.org/
RepeatMasker Genomic Datasets: http://repeatmasker.org/genomicDatasets/RMGenomicDatasets.html
TEToolkit: https://github.com/mhammell-laboratory/TEtranscripts
TEhub (collaborative platform and reference site): https://tehub.org/